

BEST-IN-CLASS CYBER SECURITY OPERATIONS PLATFORM

Full Software Requirements Specification (SRS)

SOCaaS | MDR | Managed XDR | Sovereign SOC | Enterprise Private SOC | MSSP White-Label

Prepared for: FS S / Platform Venture Planning

Version: 1.0 - Full Requirements Draft

Date: 07 July 2026

Classification: Confidential - Build Planning

Document Control

Item	Detail
Document purpose	This SRS defines the complete functional, non-functional, security, operational, data, integration, AI, compliance, and delivery requirements for a best-in-class Cyber Security Operations Platform.
Truth position	This document treats previous proposal material as a market/use-case reference only. The platform scope is intentionally broader than any single Ghana, Nigeria, UK, public-sector, financial-services, or managed-service deployment.
Intended users	Founders, product owners, Developer coding agents, senior engineers, cybersecurity architects, SOC leaders, detection engineers, cloud/DevSecOps teams, investors, procurement teams, and implementation partners.
Build approach	Use Developers as engineering accelerators, but require expert cyber architecture, senior engineering review, secure configuration, detection tuning, operational validation, and managed SOC governance before commercial launch.
Status	Comprehensive draft. Requires final validation by a senior cybersecurity architect, SOC operations lead, cloud security architect, privacy/regulatory counsel, and target-client representatives.

Revision History

Version	Date	Author/Owner	Summary
1.0	2026-07-07	ChatGPT for FS S	Full SRS created from platform vision, reference concept paper, and best-practice SOC/MDR/XDR design patterns.

Reference Frameworks and Standards Used

The SRS is aligned to widely used security operations and cyber risk references, including NIST CSF 2.0, NIST incident handling guidance, MITRE ATT&CK, OCSF, Sigma, STIX/TAXII, CIS Controls, FIRST TLP, and Zero Trust principles. These references are not copied into the product as rigid constraints; they guide taxonomy, reporting, evidence, control coverage, detection language, intelligence exchange, and operating model decisions.

Reference	Use in SRS	Source
NIST CSF 2.0	Govern, Identify, Protect, Detect, Respond, Recover outcome mapping; board/regulator reporting; cyber risk maturity views.	NIST CSWP 29, 2024; https://www.nist.gov/cyberframework
NIST SP 800-61 Rev. 2 / successor guidance	Incident handling lifecycle, triage, containment, eradication, recovery, lessons learned. Rev. 2 is archived/withdrawn by NIST, so final implementation should verify current successor guidance.	https://csrc.nist.gov/pubs/sp/800/61/r2/final
MITRE ATT&CK Enterprise	Tactic/technique mapping, detection coverage, threat-hunting library, case timeline enrichment.	https://attack.mitre.org/
OCSF	Vendor-neutral event normalization strategy for security events, entities, activities, categories, and extensions.	https://ocsf.io/
Sigma	Portable detection rule authoring format and detection-as-code governance.	https://sigmahq.io/
STIX/TAXII 2.1	Threat-intelligence objects, indicators, sightings, and exchange mechanism.	https://www.oasis-open.org/standard/taxii-version-2-1/
CIS Controls	Baseline customer control mapping, audit readiness, and managed service evidence packs.	https://www.cisecurity.org/controls
FIRST TLP 2.0	Information-sharing classification for intelligence and incident communications.	https://www.first.org/tlp/

Table of Contents - Logical Structure

1. Executive Truth Review and Scope
2. Product Vision and Objectives
3. Stakeholders, User Classes and Personas
4. Platform Operating Models and Deployment Models
5. System Context and High-Level Architecture
6. Functional Requirements by Capability Domain
7. Data, Schema, Storage and Retention Requirements
8. Integration and Connector Requirements
9. AI SOC, Detection Engineering and SOAR Requirements
10. Incident Response, SOC Operations and Service Management Requirements
11. Compliance, Regulatory Evidence and Governance Requirements
12. Non-Functional Requirements
13. Security, Privacy and Multi-Tenancy Requirements
14. User Journeys and Use Cases
15. Service Packages, Entitlements and Commercial Guardrails
16. MVP, V1, V2 and Enterprise Roadmap
17. Implementation Plan and Build Governance
18. Testing, Validation and Acceptance Criteria
19. Requirements Traceability Matrix
20. Appendices: Playbooks, Detection Use Cases, Data Dictionary, Backlog Seeds

1. Executive Truth Review and Scope

1.2 Product Scope

The system shall be a modular Cyber Security Operations Platform capable of operating as a SaaS platform, managed service platform, sovereign/local SOC platform, enterprise private SOC platform, and MSSP white-label platform. It shall provide native SOC functions while integrating with customer-owned identity, endpoint, cloud, firewall, email, vulnerability, ticketing, messaging, SIEM, and threat intelligence systems.

1.3 In Scope

- Multi-tenant SOC platform with strong tenant isolation and deployment-mode flexibility.
- Customer portal, admin portal, SOC analyst workbench, detection engineering workspace, service management workspace, and executive/regulatory reporting portal.
- Alert, log, finding, asset, identity, and vulnerability ingestion through APIs, collectors, syslog, webhooks, batch uploads, and vendor connectors.
- Correlation, deduplication, prioritisation, case management, investigation timeline, evidence handling, SLA tracking, and customer approval workflows.
- Detection-as-code library, rule testing, tuning workflow, MITRE ATT&CK coverage, and customer-specific baselines.
- SOAR playbooks with human-in-the-loop action gating, authority-to-act policy enforcement, and automated evidence capture.
- AI analyst copilot, AI investigation assistant, AI report generator, and AI detection tuning assistant with strong controls and audit logs.
- Compliance evidence modules configurable for Ghana, Nigeria, UK, financial services, public sector, critical infrastructure, data protection, and internal audit use cases.
- Operational requirements for 24/7 SOC, MDR, incident response, threat hunting, customer onboarding, customer success, and service reviews.
- Implementation roadmap, backlog seed, test strategy, and traceability matrix.

1.4 Out of Scope for Initial Build

- Replacing full enterprise EDR products such as Microsoft Defender, CrowdStrike, SentinelOne, Sophos, or equivalent endpoint agents.
- Replacing full enterprise firewalls, secure web gateways, CASB/SASE platforms, or identity providers.
- Providing unauthorised offensive security tooling, malware deployment, exploit automation, or intrusive testing without explicit customer authority.
- Autonomous destructive containment actions unless customer authority-to-act rules and human approval gates explicitly permit them.
- Guaranteeing prevention of all breaches. The platform shall provide monitoring, detection, response workflow, evidence, and risk reduction; it cannot guarantee zero incidents.

1.5 Definitions

Term	Definition
SOC	Security Operations Centre: people, process, and technology for security monitoring, detection, investigation, response, and reporting.
SOCaaS	SOC-as-a-Service: outsourced SOC capability delivered as a subscription.
MDR	Managed Detection and Response: managed service combining monitoring, investigation, detection engineering, and guided/managed response.
XDR	Extended Detection and Response: integrated detection and response across endpoint, identity, email, network, cloud, and other telemetry.
SOAR	Security Orchestration, Automation and Response: workflow automation, enrichment, action execution, and case orchestration.
Tenant	A customer, business unit, MSP client, or deployment partition with separate users, assets, logs, cases, and policies.
Authority to Act	Contractual and system policy defining which response actions the SOC may execute automatically, with approval, or not at all.
Detection as Code	Version-controlled detection rules, tests, metadata, deployment approvals, and tuning workflows.
Evidence Pack	Structured, exportable evidence set for audit, board, regulator, customer service review, or post-incident reporting.

2. Product Vision, Objectives and Success Measures

2.1 Product Vision

To build a trusted, modular, AI-assisted Cyber Security Operations Platform that enables organisations and managed security providers to detect, investigate, respond to, evidence, and report cyber threats across fragmented security toolsets, multiple customer segments, and varied regulatory environments.

2.2 Strategic Objectives

ID	Objective
OBJ-001	Provide a single operational layer for alerts, incidents, threats, assets, vulnerabilities, identities, evidence, and customer communications.
OBJ-002	Make SOCaaS and MDR commercially viable for organisations that cannot build their own 24/7 SOC.
OBJ-003	Support enterprise-grade and sovereign deployment models without rewriting the product for each country or sector.
OBJ-004	Use AI to accelerate triage, investigation, reporting, and tuning, while keeping containment decisions safe, explainable, and auditable.
OBJ-005	Create configurable service packages with clear entitlements, limits, margins, SLAs, and add-on services.
OBJ-006	Enable expert cybersecurity teams to continuously improve detections, playbooks, coverage, and customer security outcomes.
OBJ-007	Provide board, regulator, audit, and customer evidence without requiring manual spreadsheet-based SOC reporting.
OBJ-008	Deliver a product architecture that can be built in phases by engineering teams and Developers without locking the venture into one vendor stack.

2.3 Success Measures

Category	Measure	Target / Design Goal
Security Operations	Mean time to triage Priority 1 alert	< 15 minutes for managed service tiers once onboarded and tuned.
Security Operations	Mean time to customer notification	Within tier-specific SLA and customer communication policy.
Detection Quality	False-positive rate after tuning period	< 20% per tenant for mature rules; monitored by rule and data source.
Platform Reliability	Core platform uptime	99.9% MVP, 99.95% enterprise tier, 99.99% optional dedicated/critical tier.
Data Pipeline	Ingestion delay	Near-real-time for streaming connectors; target < 2 minutes for supported APIs/syslog where source permits.
AI Safety	Ungated high-impact action rate	0. No destructive or business-impacting response action without appropriate policy and approval.
Commercial	Gross margin protection	Package limits and overage billing enforced through system entitlements.
Compliance	Evidence pack availability	100% on-time for regulated tenants with configured compliance templates.
Customer Trust	Auditability	All analyst, system, AI, and automation decisions must be traceable.

3. Stakeholders, User Classes and Personas

3.1 Stakeholder Groups

Stakeholder	Primary Interest	Platform Implications
Customer CISO / Security Lead	Visibility, response confidence, reporting, evidence, SLA performance.	Executive dashboards, severity transparency, service review packs, approved playbooks, data source health.
Customer IT Admin	Connector setup, remediation tasks, asset context, user account actions.	Guided onboarding, least-privilege integration setup, approval workflows, action audit trail.
SOC Tier 1 Analyst	Fast triage, enrichment, deduplication, clear escalation.	Alert queue, triage forms, AI summary, runbook steps, escalation rules.
SOC Tier 2 Analyst	Deep investigation, timeline, correlated evidence.	Entity graph, raw event access, related alerts, investigation notebook, customer context.
SOC Tier 3 / IR Lead	Major incident command, containment, eradication, forensic evidence.	War room, high-risk action controls, evidence chain, post-incident report.
Detection Engineer	Rule lifecycle, test data, coverage, tuning.	Detection-as-code repository, rule lab, MITRE coverage, deployment workflow.
Threat Intelligence Analyst	Indicators, campaigns, advisories, intelligence sharing.	STIX/TAXII feeds, indicator lifecycle, intel-to-detection workflow, TLP labels.
MSSP Operator	Many customers, white-labeling, entitlements, billing, cross-tenant benchmarking.	Partner hierarchy, tenant templates, package limits, MSP dashboards.
Regulator / Auditor	Evidence, incident timelines, control coverage, governance.	Exportable evidence packs, immutable logs, regulatory dashboards.
Investor / Executive Board	ARR, margin, operational maturity, risk management.	Business KPIs, utilisation, churn, incident trends, audit outcomes.

3.2 User Roles

Role	Description
Platform Super Admin	Internal highest-privilege role; configures global system settings, deployment settings, tenant creation, platform health, billing defaults, and emergency controls.
Tenant Admin	Customer-side administrator; manages tenant users, integrations, assets, approval policies, reporting recipients, and compliance settings.
Customer Executive Viewer	Read-only customer role for board-level dashboards, risk trends, SLA reports, and evidence packs.
Customer Technical User	Customer IT/security user who can view alerts, approve actions, provide evidence, and execute customer-side tasks.
SOC Manager	Internal operations lead; manages queues, analyst assignments, quality review, SLAs, shift handover, and escalation.
Tier 1 Analyst	Performs initial triage, enrichment review, false-positive closure, escalation and customer notifications per runbooks.
Tier 2 Analyst	Performs deeper investigation, timeline building, containment recommendations, detection feedback, and incident coordination.
Tier 3 Analyst / Incident Responder	Handles severe incidents, threat hunting, forensic readiness, major containment actions, and post-incident reviews.
Detection Engineer	Creates, tests, deploys, tunes and retires detection content and correlation rules.
Threat Intel Analyst	Manages intel feeds, indicators, advisories, intelligence confidence, TLP classifications, and campaign mapping.
Compliance Manager	Manages regulatory templates, evidence packs, audit requests, policy mapping, and customer assurance outputs.
MSSP Partner Admin	Manages downstream tenants, package templates, white-label settings, partner analysts, and partner reporting.
API Client	Machine role used for integrations, collectors, automation, and customer API access with scoped permissions.

4. Platform Operating Models and Deployment Models

4.1 Supported Commercial/Operating Models

Model	Description	Key Requirements
SOCaaS	Provider-operated SOC monitoring service for customers that lack internal SOC capability.	Multi-tenant portal, 24/7 queue, clear packages, onboarding factory, SLA evidence.
MDR	Provider-operated detection and response with deeper investigation and guided/managed containment.	EDR/XDR integrations, threat intel, response playbooks, authority-to-act controls.
Managed XDR	Cross-domain detection and response across endpoint, identity, network, email, cloud, vulnerability and threat intel.	Correlation engine, normalized schema, entity graph, high-quality connectors.
Sovereign SOC	Country/local hosted SOC for government, CII, financial services or regulated sectors.	Data residency, local compliance pack, dedicated hosting option, local governance.
Enterprise Private SOC	Dedicated platform instance for one enterprise with optional managed service support.	Dedicated tenancy, SSO, private connectors, custom retention, integration with customer SIEM.
MSSP White-Label	Platform used by managed service partners to serve their clients under their brand.	Partner hierarchy, branding, downstream tenants, reseller billing and analyst partitions.
Hybrid Co-Managed SOC	Customer and provider jointly operate detections and response.	Shared queues, ownership rules, co-managed playbooks, task assignments, split SLAs.

4.2 Deployment Models

Deployment Model	When Used	Requirements
Shared SaaS	SMEs, mid-market, standard SOCaaS customers, fast onboarding.	Strong logical tenant isolation, pooled scalable services, package-based limits.
Dedicated SaaS Tenant	Regulated enterprise requiring more isolation but still managed cloud.	Dedicated data stores or namespaces, dedicated keys, advanced logging, configurable retention.
Sovereign Cloud / Local Country Hosting	Government, CII, data residency, national/regional SOC propositions.	Country-specific hosting, legal entity mapping, local backup/DR, local access policy.
Customer Private Cloud	Enterprise/private SOC customers with internal cloud preference.	Deployable stack, infrastructure-as-code, customer-managed network, secure update model.
On-Prem / Air-Gapped Adjacent Collector	Highly sensitive networks or OT environments.	Local collector, buffering, one-way export options, minimal dependencies, strict update controls.
Hybrid Collector Model	Most enterprise deployments.	Customer-side collectors normalize/filter/forward telemetry to platform; raw data residency configurable.

4.3 Product Principles

- Native core SOC functions must work even when only basic syslog/API/webhook feeds exist.
- Integrations increase value but the product must not require every customer to have every tool.
- Tenant isolation, evidence integrity, and auditability are foundational; they are not premium extras.
- AI must accelerate analysts but cannot become an uncontrolled authority for destructive actions.
- The product must support package limits and commercial guardrails from day one.
- Every incident, detection, playbook, and customer communication must be traceable to data and decision history.
- The architecture must allow vendor substitution: the platform may integrate with Microsoft, CrowdStrike, SentinelOne, Fortinet, Palo Alto, and others without becoming dependent on one ecosystem.

5. System Context and High-Level Architecture

5.1 System Context

The platform sits above customer tools and provider operations. It ingests alerts, logs, findings, identities, assets, vulnerabilities, and threat intelligence. It normalizes and enriches data, correlates signals, creates incidents, guides analysts, automates approved actions, communicates with customers, and generates evidence and reports.

External Source / Sink	Direction	Examples	Primary Use
Identity providers	Inbound and outbound	Entra ID, Okta, Google Workspace	Identity alerts, risky sign-ins, user context, disable/revoke-session actions.
Endpoint security	Inbound and outbound	Microsoft Defender, CrowdStrike, SentinelOne, Sophos	EDR alerts, device context, isolate host, collect investigation package.
Network/firewall/SASE	Inbound and outbound	Fortinet, Palo Alto, Cisco, Zscaler, Cloudflare	Network detections, blocked traffic, malicious IP/domain actions.
Cloud platforms	Inbound and outbound	AWS, Azure, GCP	CloudTrail/activity logs, IAM changes, GuardDuty/Security Command Center alerts, cloud response tasks.
SIEM/data lakes	Inbound and outbound	Microsoft Sentinel, Splunk, Elastic/OpenSearch, BigQuery	Alert import/export, query delegation, log search, customer existing investment leverage.
Ticketing/ITSM	Outbound and inbound	ServiceNow, Jira, email	Ticket creation, status sync, remediation task tracking.
Messaging	Outbound and inbound	Teams, Slack, email, SMS/WhatsApp gateways	Notifications, approvals, war-room comms, customer escalation.
Threat intelligence	Inbound and outbound	STIX/TAXII, MISP, OTX, commercial feeds	Indicator enrichment, campaign context, detection updates.
Vulnerability tools	Inbound	Tenable, Qualys, Rapid7, Defender Vulnerability Management	Exposure prioritization and incident context.
Customer APIs	Inbound and outbound	Custom business systems, sector platforms	Sector-specific telemetry, evidence, regulatory workflows.

5.2 Logical Architecture

The logical architecture shall be divided into presentation, API, identity, tenant management, ingestion, normalization, storage, analytics, detection, correlation, case management, SOAR, AI, reporting, integration, audit, and observability layers. Each layer shall expose clear APIs and event contracts to avoid tight coupling.

Layer	Core Components	Key Requirements
Presentation	Customer portal, analyst workbench, admin portal, executive dashboard, mobile-responsive views.	Role-based navigation, tenant isolation, accessible UI, fast search, export controls.
API Gateway	REST/GraphQL APIs, webhook endpoints, collector APIs, partner APIs.	Authentication, rate limits, idempotency, schema validation, request logging.
Identity & Access	SSO, MFA, RBAC/ABAC, service accounts, API tokens.	Least privilege, tenant boundaries, privileged access review, break-glass controls.
Ingestion	Connectors, syslog collectors, agent/collector, batch import, queue/broker.	Scalable, resumable, source-health monitoring, deduplication, backpressure.
Normalization	Parser framework, OCSF-inspired schema, enrichment pipeline.	Versioned mappings, parser tests, unknown-field preservation.
Storage	Hot event store, incident DB, evidence object store, analytics warehouse, archive.	Encryption, retention policies, query performance, tenant partitioning.
Detection & Correlation	Rule engine, Sigma-like rules, behavioral rules, risk scoring, correlation graph.	Versioning, tests, customer tuning, MITRE mapping, false-positive feedback.
Case Management	Alerts, incidents, tasks, notes, timelines, evidence, SLAs.	Full lifecycle, audit logs, customer visibility controls, assignment workflow.
SOAR	Playbook designer, enrichment steps, action catalog, approval gates.	Safe execution, rollback guidance, authority-to-act enforcement.
AI Layer	AI triage, investigation, summarization, reporting, tuning assistant.	Grounded outputs, citations to evidence, prompt logging, model risk controls.
Reporting	Operational, executive, compliance, service review, regulatory exports.	Templates, scheduling, watermarking, export audit, customer branding.
Observability	Metrics, logs, traces, health checks, synthetic tests.	SLOs, alerting, capacity forecasts, ingestion health.

5.3 Recommended Technology Direction

The SRS does not mandate a single stack, but the recommended build pattern is cloud-native, API-first, event-driven, and modular. A likely stack is Next.js/React for portals; Go, Java/Kotlin, or Node/NestJS for services; PostgreSQL for relational data; OpenSearch/ClickHouse/Elastic/Snowflake/BigQuery depending on deployment for event analytics; Kafka/Redpanda/NATS for streaming; Temporal or equivalent for workflows; object storage for evidence; Vault/cloud secrets manager for secrets; Kubernetes or managed containers for deployment; OpenTelemetry/Prometheus/Grafana for observability; and Terraform/OpenTofu for infrastructure as code.

6. Functional Requirements by Capability Domain

6.1 Tenant, Organisation and Account Management

Req ID	Requirement	Priority	Phase	Acceptance Evidence
TEN-001	The system shall support hierarchical tenant structures including parent customer, subsidiary, department, business unit, MSSP partner, downstream customer, and dedicated sovereign programme.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-002	The system shall isolate users, logs, assets, incidents, evidence, dashboards, reports, billing records, configurations, and API credentials by tenant.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-003	The system shall support tenant templates for onboarding common customer types such as SME, bank, telecom, utility, public-sector agency, healthcare provider, fintech, and MSSP customer.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-004	The system shall support tenant status lifecycle: prospect, onboarding, active, suspended, churned, archived, and legally held.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-005	The system shall support tenant-level package entitlement: asset limits, log volume limits, connectors, retention, SLA tier, included reports, playbook permissions, IR retainer hours, and AI features.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-006	The system shall allow tenant admins to maintain organisation profile, contacts, escalation matrix, business hours, critical assets, legal/regulatory profile, and authority-to-act policy.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-007	The system shall prevent cross-tenant data leakage through database design, API checks, search filters, reporting engine controls, cache scoping, and audit controls.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-008	The system shall support customer-specific branding and white-labeling for authorised MSSP partners.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-009	The system shall support tenant offboarding, export, retention freeze, archive, deletion workflow, legal hold, and certificate of destruction where permitted.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TEN-010	The system shall maintain a tenant change history for all material settings changes.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Tenant, Organisation and Account Management: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.2 Identity, Access Management and Session Security

Req ID	Requirement	Priority	Phase	Acceptance Evidence
IAM-001	The system shall support local users, SSO through SAML/OIDC, MFA, API keys, service accounts, and temporary invitation links.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-002	The system shall implement role-based access control with optional attribute-based constraints for tenant, business unit, case sensitivity, data source, geography, and role duty.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-003	The system shall require MFA for internal platform users, SOC analysts, tenant admins, and any user with approval authority or export rights.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-004	The system shall support privileged access management workflows for platform super admins, including justification, time-bounded elevation, logging, and approval for high-risk actions.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-005	The system shall enforce least-privilege API scopes and token rotation for connector credentials.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-006	The system shall support break-glass access with emergency reason capture, automatic alerting, and post-use review.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-007	The system shall provide configurable session timeout, IP restrictions, device trust indicators, and geo-anomaly logging.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-008	The system shall provide user lifecycle events including invitation, activation, suspension, role change, password reset, MFA reset, and deactivation.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-009	The system shall provide access review reports for customer admins and internal compliance.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
IAM-010	The system shall log failed login attempts, MFA failures, suspicious admin behaviour, token usage, export access, and unusual session patterns.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Identity, Access Management and Session Security: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.3 User Experience, Portals and Dashboards

Req ID	Requirement	Priority	Phase	Acceptance Evidence
UI-001	The system shall provide distinct but coherent experiences for customer users, SOC analysts, detection engineers, threat intelligence analysts, compliance users, platform admins, and MSSP partners.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
UI-002	The customer portal shall show incidents, risk posture, assets, integration health, open tasks, approvals, service metrics, reports, and evidence packs.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
UI-003	The analyst workbench shall show priority queues, alert context, entity graph, timeline, enrichment, runbook steps, AI assistance, notes, and escalation options.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
UI-004	The executive dashboard shall show risk trends, critical incidents, SLA compliance, exposure trends, control coverage, remediation progress, and business impact language.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
UI-005	The admin portal shall support tenant setup, entitlements, connector configuration, user roles, platform settings, feature flags, and billing.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
UI-006	The detection engineering workspace shall support rule authoring, testing, deployment, version history, tuning feedback, MITRE mapping, and coverage dashboards.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Confidential - Draft for build planning and expert validation

UI-007	The compliance portal shall support audit requests, evidence packs, reporting templates, regulatory calendars, and control mappings.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
UI-008	The UI shall be responsive for tablet/laptop use and support dark mode for analyst SOC displays.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
UI-009	The UI shall include clear warning states for high-risk response actions and AI-generated recommendations.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
UI-010	The UI shall meet accessibility expectations for colour contrast, keyboard navigation, labels, and readable dashboards.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for User Experience, Portals and Dashboards: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.4 Data Ingestion, Collection and Connector Framework

Req ID	Requirement	Priority	Phase	Acceptance Evidence
ING-001	The system shall ingest alerts, events, logs, assets, identities, vulnerabilities, tickets, threat intelligence, and customer metadata through connectors, syslog, webhooks, batch uploads, and APIs.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-002	The system shall provide a connector SDK with standard patterns for authentication, pagination, incremental polling, checkpointing, retry, error handling, normalization, and health metrics.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-003	The system shall support customer-side collectors for syslog, Windows event forwarding, cloud APIs, local file pickup, OT-adjacent feeds, and buffered offline operation.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-004	The system shall validate all inbound data against source schema and platform canonical schema, preserving raw original records in accordance with retention policy.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-005	The system shall detect ingestion gaps, connector failures, authentication failures, rate-limit issues, schema drift, duplicate payloads, and abnormal data volume changes.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-006	The system shall support backpressure and durable queues to prevent data loss under burst conditions.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-007	The system shall support tenant-specific routing and encryption from ingestion edge to storage.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-008	The system shall support source priority and classification so that critical data sources affect SLA and health dashboards.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-009	The system shall allow replay of historical data for testing new detections where retention permits.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ING-010	The system shall support ingestion quotas, overage tracking, package enforcement, and commercial alerts when customers exceed contracted volumes.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Data Ingestion, Collection and Connector Framework: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.5 Normalization, Enrichment and Entity Resolution

Req ID	Requirement	Priority	Phase	Acceptance Evidence
NORM-001	The system shall normalize ingested data into a canonical event model inspired by OCSF-style categories, activities, observables, actors, assets, identities, network entities, cloud resources, and finding classes.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-002	The system shall preserve raw source fields for forensic traceability while exposing normalized fields for detection, search, correlation, dashboards, and reports.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-003	The system shall support parser versioning, regression tests, field mapping documentation, and schema drift alerts.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-004	The system shall enrich events with tenant context, asset criticality, user role, business unit, geo-location, threat intelligence, vulnerability exposure, and recent incident history.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-005	The system shall resolve entities across aliases, hostnames, IPs, MAC addresses, cloud instance IDs, user principal names, email addresses, device IDs, and external IDs.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-006	The system shall distinguish confidence levels in entity resolution and show analysts when relationships are inferred versus authoritative.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-007	The system shall support enrichment cache TTLs and prevent stale enrichment from misleading incident decisions.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-008	The system shall support customer-specific asset importance, crown-jewel tagging, and critical business-service mapping.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-009	The system shall support data quality dashboards showing unmapped fields, missing required fields, parser failure rates, and enrichment coverage.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
NORM-010	The system shall support privacy-preserving redaction or masking rules for sensitive fields by tenant and user role.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Normalization, Enrichment and Entity Resolution: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.6 Detection Engineering and Rules Management

Req ID	Requirement	Priority	Phase	Acceptance Evidence
DET-001	The system shall provide a detection-as-code framework for rules, metadata, tests, dependencies, owners, version history, deployment workflow, and rollback.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-002	The system shall support Sigma-like detection authoring and translation where applicable, plus native behavioral, threshold, sequence, correlation, and anomaly rules.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-003	Every detection shall include title, description, severity, confidence, ATT&CK mapping, data source requirements, false-positive guidance, response playbook, owner, status, and test cases.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-004	The system shall support global rules, industry pack rules, tenant-specific rules, beta rules, disabled rules, and emergency rules.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-005	The system shall provide rule testing against sample data and historical tenant data where permitted.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-006	The system shall support promotion flow: draft, peer review, QA test, pilot, production, tuned, deprecated, retired.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-007	The system shall capture analyst feedback and false-positive dispositions to inform rule tuning.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-008	The system shall provide MITRE ATT&CK coverage dashboards by tenant, package, sector, data source, and detection confidence.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-009	The system shall detect and warn when a rule depends on a data source not active for a tenant.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
DET-010	The system shall support emergency deployment of high-priority detections with post-deployment review.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Detection Engineering and Rules Management: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.7 Alert Correlation, Risk Scoring and Prioritisation

Req ID	Requirement	Priority	Phase	Acceptance Evidence
COR-001	The system shall deduplicate repeated alerts from the same source and correlate related alerts across endpoint, identity, email, network, cloud, vulnerability, and threat intelligence data.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-002	The system shall compute alert and incident risk scores using severity, confidence, asset criticality, user privilege, threat intel confidence, vulnerability exposure, business impact, recurrence, and kill-chain progression.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-003	The system shall support configurable correlation windows and tenant-specific correlation rules.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-004	The system shall group alerts into incidents where they share meaningful entities, sequence, campaign, threat actor, malware family, user account, host, domain, IP, or business process.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-005	The system shall identify escalation triggers such as privileged account involvement, crown-jewel asset, ransomware indicator, lateral movement, data exfiltration, persistence, or active exploitation.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-006	The system shall maintain explainability for risk scores by showing contributing factors and evidence.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-007	The system shall support suppression windows, maintenance windows, and approved-change context to reduce noise.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-008	The system shall identify alert storms and allow queue-level incident command mode.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-009	The system shall provide analyst-adjustable severity with mandatory reason capture.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COR-010	The system shall continuously measure correlation effectiveness and over-correlation risk.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Alert Correlation, Risk Scoring and Prioritisation: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.8 Alert, Incident and Case Management

Req ID	Requirement	Priority	Phase	Acceptance Evidence
CASE-001	The system shall manage alert lifecycle: new, enriched, triaged, escalated, suppressed, converted to incident, duplicate, false positive, benign true positive, closed.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-002	The system shall manage incident lifecycle: opened, assigned, investigating, waiting customer, containment pending, contained, eradication, recovery, monitoring, closed, post-incident review.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-003	Every case shall include unique ID, tenant, title, severity, priority, category, affected entities, source alerts, timeline, owner, SLA, status, tasks, notes, evidence, communications, and audit trail.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-004	The system shall support internal-only and customer-visible notes, evidence, tasks, and status updates.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-005	The system shall support task assignment to analysts, customer users, external partners, or automated playbooks.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-006	The system shall support parent-child incidents, related incidents, duplicates, and major incident aggregation.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-007	The system shall provide templates for incident categories such as phishing, BEC, malware, ransomware, endpoint compromise, identity compromise, cloud compromise, data exfiltration, vulnerability exploitation, insider threat, DDoS, and policy violation.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-008	The system shall support attaching files, screenshots, log extracts, hashes, URLs, email headers, forensic artifacts, and chain-of-custody metadata.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-009	The system shall enforce closure criteria including disposition, root cause, impact, actions taken, lessons learned, and customer acknowledgement where required.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
CASE-010	The system shall provide searchable case history and knowledge-base linking.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Alert, Incident and Case Management: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.9 Investigation Workbench, Timeline and Entity Graph

Req ID	Requirement	Priority	Phase	Acceptance Evidence
INV-001	The system shall provide a unified investigation view with correlated alerts, raw events, entity profile, enrichment, timeline, runbook, notes, tasks, and AI summary.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
INV-002	The timeline shall show event time, ingest time, source, entity, action, severity, confidence, analyst actions, automation actions, customer communications, and evidence capture.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
INV-003	The entity graph shall show relationships among users, devices, IPs, domains, files, processes, emails, cloud resources, vulnerabilities, tickets, and incidents.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
INV-004	The system shall allow analysts to pivot from an entity to related events, incidents, vulnerabilities, alerts, and historical activity.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
INV-005	The system shall support investigation notebooks with structured sections for hypothesis, evidence, analysis, decision, and next action.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
INV-006	The system shall support query builder and advanced search across normalized fields and raw data, subject to user permissions and retention.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
INV-007	The system shall preserve every query, export, and evidence selection in the audit log for	Must	MVP/V1 depending on dependency	Functional test +

	sensitive cases.		on dependency	role/tenant security test
INV-008	The system shall support saved investigation views and shareable internal links.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
INV-009	The system shall show missing context and data-source gaps to prevent false confidence.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
INV-010	The system shall support war-room mode for major incidents with live timeline updates and assigned roles.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Investigation Workbench, Timeline and Entity Graph: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.10 Threat Intelligence Management

Req ID	Requirement	Priority	Phase	Acceptance Evidence
TI-001	The system shall ingest threat intelligence through STIX/TAXII, MISP-like feeds, CSV/API imports, commercial feeds, internal advisories, and manual analyst submissions.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-002	The system shall support intelligence objects including indicators, malware, campaigns, threat actors, attack patterns, tools, vulnerabilities, reports, and sightings.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-003	The system shall track confidence, source reliability, first seen, last seen, TLP classification, expiry, kill-chain phase, ATT&CK mapping, and permitted sharing scope.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-004	The system shall enrich alerts and incidents with relevant intelligence and show why a match matters.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-005	The system shall prevent blind blocking on low-confidence indicators without analyst or policy review.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-006	The system shall support converting intelligence into detection rules, watchlists, hunting queries, customer advisories, and evidence-pack references.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-007	The system shall support sector-specific intelligence packs and country/regional intelligence packs.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-008	The system shall support indicator decay, expiry, revocation, and suppression to reduce stale intelligence risk.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-009	The system shall support internal intelligence reporting and customer-facing advisories with TLP labels.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
TI-010	The system shall support intelligence sharing communities where legally and contractually permitted.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Threat Intelligence Management: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.11 SOAR, Playbooks and Response Automation

Req ID	Requirement	Priority	Phase	Acceptance Evidence
SOAR-001	The system shall provide a playbook engine with trigger, condition, enrichment, decision, approval, action, notification, wait, branch, rollback-guidance, and closure steps.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-002	The system shall provide a controlled action catalog including notify, ticket, enrich, block IP/domain/hash, quarantine email, disable user, revoke session, isolate endpoint, collect evidence, request customer action, and generate report.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-003	The system shall enforce authority-to-act rules per tenant, package, data source, action type, severity, business hours, and approval group.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-004	The system shall classify actions as informational, low impact, medium impact, high impact, and business critical.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-005	The system shall require human approval for high-impact actions unless contractually pre-authorised and technically safe.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-006	The system shall log every playbook execution, input, output, decision, approval, error, retry, and action result.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-007	The system shall support dry-run/simulation mode for playbook testing.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-008	The system shall support playbook versioning, review, approval, rollback, and tenant-specific overrides.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-009	The system shall support playbook failure handling, partial execution recovery, and escalation to human analysts.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
SOAR-010	The system shall support business continuity mode where automation is reduced during known integration instability.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for SOAR, Playbooks and Response Automation: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.12 AI SOC Copilot and Agentic Investigation

Req ID	Requirement	Priority	Phase	Acceptance Evidence
AI-001	The system shall provide AI assistance for triage summaries, incident narratives, evidence extraction, timeline explanation, probable root cause, recommended next steps, and report drafts.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-002	AI outputs shall be grounded in retrieved platform evidence and cite the underlying alerts, events, entities, case notes, and playbook steps used.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-003	AI shall distinguish facts, inferences, assumptions, and recommended actions.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-004	AI shall not execute containment or destructive actions directly; it may recommend or prepare actions for approved playbooks subject to authority-to-act controls.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-005	The system shall log prompts, context packages, model outputs, user edits, approvals, rejections, and feedback for audit and model evaluation.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-006	The system shall prevent tenant data from being used to train public models unless explicitly contracted and technically controlled.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-007	The system shall support model selection by deployment model, including private model, hosted model, customer-managed model, or no-AI mode.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-008	The system shall support AI safety tests for hallucination, unsafe recommendation, cross-tenant leakage, prompt injection, and unsupported claims.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-009	The system shall support redaction and minimisation of sensitive fields before sending data to external AI services where allowed.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
AI-010	The system shall present AI assistance as decision support, not a final authority.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for AI SOC Copilot and Agentic Investigation: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.13 Reporting, Dashboards and Evidence Packs

Req ID	Requirement	Priority	Phase	Acceptance Evidence
REP-001	The system shall generate operational, customer, executive, board, audit, regulatory, service review, and post-incident reports.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-002	The reporting engine shall support templates, scheduled delivery, ad hoc export, filtering, watermarking, approval, and distribution lists.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-003	Reports shall include clear data-period, tenant, scope, data-source coverage, limitations, definitions, and evidence references.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-004	The system shall generate monthly service review packs including incidents, SLA performance, false-positive rate, top risks, integrations health, actions taken, recommendations, and open remediation tasks.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-005	The system shall generate executive cyber posture summaries in business language.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-006	The system shall generate regulatory evidence packs aligned to configured frameworks and jurisdiction-specific templates.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-007	The system shall provide export formats including PDF, DOCX, XLSX/CSV, JSON, and API where appropriate.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-008	The system shall audit every report generation, export, and download.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-009	The system shall prevent users from exporting data outside their permissions or tenant scope.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
REP-010	The system shall support white-label reporting for MSSP partners.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Reporting, Dashboards and Evidence Packs: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.14 Compliance, Control Mapping and Regulatory Evidence

Req ID	Requirement	Priority	Phase	Acceptance Evidence
COMP-001	The system shall map detections, incidents, evidence, response actions, assets, vulnerabilities, and reports to configurable control frameworks.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-002	The system shall provide framework templates for NIST CSF, CIS Controls, ISO 27001-style control domains, data protection, critical infrastructure, and customer-specific regulatory obligations.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-003	The system shall support Ghana sovereign SOC and similar country-specific packs as configurable market packages rather than hardcoded product logic.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-004	The system shall support evidence requests, evidence owner assignment, due dates, status tracking, and exportable evidence bundles.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-005	The system shall maintain chain-of-custody metadata for incident evidence and forensic artifacts.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-006	The system shall support regulatory incident notification preparation, including timeline, affected systems, known impact, actions taken, and next steps.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-007	The system shall support policy acknowledgement and customer shared-responsibility documentation.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-008	The system shall support audit-readiness dashboards showing control coverage and evidence freshness.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-009	The system shall support retention and deletion rules by jurisdiction, contract, tenant, and data class.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMP-010	The system shall support compliance exceptions and risk acceptance records.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Compliance, Control Mapping and Regulatory Evidence: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.15 Asset, Identity, Vulnerability and Exposure Management

Req ID	Requirement	Priority	Phase	Acceptance Evidence
ASSET-001	The system shall maintain an asset inventory from connectors, manual uploads, agent/collector feeds, cloud inventories, vulnerability tools, and customer APIs.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-002	Assets shall include type, owner, business unit, criticality, location, environment, operating system, cloud account, exposure status, tags, vulnerabilities, and related incidents.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-003	The system shall maintain identity inventory including users, service accounts, privileged accounts, groups, roles, status, MFA state where available, and risky sign-in context.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-004	The system shall ingest vulnerabilities and map them to assets, criticality, exploitability, active exploitation intelligence, and open remediation tasks.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-005	The system shall support crown-jewel asset tagging and business service mapping.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-006	The system shall support attack surface and exposure dashboards for managed vulnerability add-on services.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-007	The system shall correlate alerts with exposed vulnerabilities and privileged identities to increase priority.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-008	The system shall support exceptions, accepted risk, compensating controls, and remediation target dates.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-009	The system shall support asset confidence scoring when data comes from multiple conflicting sources.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ASSET-010	The system shall support customer tasks for missing owner, missing criticality, stale asset, and unresolved vulnerability context.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Asset, Identity, Vulnerability and Exposure Management: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.16 Notification, Communication and Collaboration

Req ID	Requirement	Priority	Phase	Acceptance Evidence
COMM-001	The system shall support email, Teams, Slack, SMS/WhatsApp gateway integration where permitted, in-platform notifications, API callbacks, and ITSM ticket comments.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-002	The system shall route communications according to tenant escalation matrix, severity, business hours, package SLA, and incident type.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-003	The system shall support customer approval requests for response actions with clear risk, evidence, recommended action, expiry, and fallback path.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-004	The system shall support internal analyst shift handover notes and major incident war-room channels.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-005	The system shall maintain immutable communication logs for incident cases.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-006	The system shall support notification throttling, digest mode, and escalation if no acknowledgement is received.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-007	The system shall support different communication templates for technical users, executives, regulators, and auditors.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-008	The system shall support bilingual/localized templates for country deployments where required.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-009	The system shall support secure links with expiry for sensitive reports and approval actions.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
COMM-010	The system shall prevent sensitive incident details from being sent to insecure channels unless explicitly allowed by policy.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Notification, Communication and Collaboration: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.17 Packages, Entitlements, Billing and Commercial Controls

Req ID	Requirement	Priority	Phase	Acceptance Evidence
BILL-001	The system shall define commercial packages with included assets, users, connectors, log volume, retention, SLA, reports, AI features, playbooks, and analyst entitlements.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
BILL-002	The system shall meter log volume, alert volume, storage, connector count, asset count, report count, API usage, playbook actions, and professional service hours.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
BILL-003	The system shall support overage alerts, contract thresholds, billing export, and finance review workflows.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
BILL-004	The system shall support add-on services including incident response, digital forensics, vulnerability scanning, tabletop exercises, compliance evidence packs, threat hunting, and custom integrations.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
BILL-005	The system shall prevent silent scope creep by requiring commercial approval for unsupported custom work.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
BILL-006	The system shall support customer renewal status, contract dates, payment terms, service suspension, and downgrade/upgrade workflows.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
BILL-007	The system shall support partner/reseller pricing and downstream tenant billing metadata.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
BILL-008	The system shall support margin dashboards by package, customer, connector, storage volume, analyst time, and add-on service.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
BILL-009	The system shall support onboarding fees, mobilisation fees, minimum annual commitments, and framework/anchor customer pricing rules.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

BILL-010	The system shall integrate with external finance systems in later phases.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
----------	---	------	--------------------------------	---

Implementation note for Packages, Entitlements, Billing and Commercial Controls: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

6.18 Platform Administration, Configuration and Feature Flags

Req ID	Requirement	Priority	Phase	Acceptance Evidence
ADMIN-001	The system shall provide global configuration for packages, connectors, rule packs, report templates, retention policies, AI providers, action catalog, and deployment settings.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-002	The system shall support feature flags by environment, tenant, package, partner, region, and beta programme.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-003	The system shall provide platform health dashboards for queues, storage, API latency, connector health, job failures, AI service health, and report generation status.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-004	The system shall support configuration audit logs and rollback for material settings.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-005	The system shall provide admin workflows for tenant creation, migration, suspension, export, legal hold, and deletion.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-006	The system shall provide data repair and reprocessing tools with strict permissions and audit logs.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-007	The system shall support content library management for detections, playbooks, report templates, compliance mappings, and knowledge base articles.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-008	The system shall support maintenance windows, incident banners, release notes, and customer-visible platform status.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-009	The system shall support secrets rotation reminders and connector credential expiry tracking.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test
ADMIN-010	The system shall support support-ticket and customer support workflows for platform issues.	Must	MVP/V1 depending on dependency	Functional test + role/tenant security test

Implementation note for Platform Administration, Configuration and Feature Flags: the build team shall convert each requirement above into user stories, API contracts, database migrations, UI screens, validation tests, permission tests, and operational test scripts. Cybersecurity subject matter experts shall review security-sensitive workflows before production release.

7. Data, Schema, Storage and Retention Requirements

7.1 Canonical Data Domains

Data Entity	Description
Tenant	Organisation/customer partition with package, settings, contacts, policies, compliance profile, and deployment metadata.
User	Platform user or customer user with identity, roles, permissions, MFA status, tenant association, and audit history.
Role/Permission	Access-control configuration including RBAC/ABAC constraints, action permissions, export rights, and approval authority.
Integration	Configured connector with credentials, scopes, status, health, data types, polling settings, and source metadata.
Collector	Customer-side or platform-side data collection service with version, host, route, buffer status, and health.
Raw Event	Original event/log/alert/finding payload preserved in source format subject to retention and privacy policy.
Normalized Event	Canonical representation of a source event with mapped fields, metadata, confidence, and entity references.
Alert	Actionable security signal generated by source tool, detection rule, correlation rule, AI-assisted review, or manual submission.
Incident/Case	Investigation container with alerts, timeline, entities, tasks, evidence, notes, communications, SLAs, and closure data.
Entity	User, host, IP, domain, URL, file, process, cloud resource, email, account, vulnerability, or business service involved in events.
Asset	Device, server, workload, cloud resource, application, OT/IoT system, network device, or business system under monitoring.
Vulnerability	Weakness or exposure mapped to assets, severity, exploitability, remediation state, and related incidents.
Detection Rule	Versioned logic, metadata, test cases, status, mapping, and tuning information.
Playbook	Versioned workflow with trigger, conditions, steps, approvals, actions, and execution logs.
Threat Intelligence Object	Indicator, campaign, actor, tool, malware, attack pattern, report, sighting, or intelligence note.
Evidence Item	Artifact attached to case/report with source, hash, chain-of-custody, classification, and retention.
Report	Generated document/dashboard export with scope, template, recipients, approvals, and access history.
Audit Log	Immutable record of user, system, API, AI, automation, admin, export, and data-access events.
Billing Usage Record	Metered record of assets, logs, connectors, storage, reports, playbook actions, AI usage, and service hours.
Compliance Evidence Record	Control-mapped evidence, owner, status, expiry, exception, risk acceptance, and export history.

7.2 Canonical Event Field Requirements

Field	Requirement
event_id	Unique platform event ID.
tenant_id	Tenant partition ID; mandatory for every record.
source_system	Connector/source tool name and instance.
source_event_id	Original source event ID if available.
event_time	Time event occurred at source.
ingest_time	Time platform received event.
event_category	Normalized category such as identity, endpoint, network, email, cloud, vulnerability, threat intel, system.
event_activity	Normalized activity/action.
severity	Source severity and normalized severity.
confidence	Confidence in detection or enrichment.
actor	User, process, service, or external actor.
target	Affected resource, user, file, or asset.
src_ip	Source IP.
dst_ip	Destination IP.
domain	Domain/hostname if applicable.
url	URL if applicable.
file_hash	Hash fields for file observables.
process	Process name/path/cmdline if applicable.
cloud_account	Cloud tenant/account/subscription/project.
asset_id	Resolved platform asset ID.
identity_id	Resolved identity ID.
raw_ref	Pointer to raw event payload.
parser_version	Parser version used.
data_classification	Classification of event fields and masking rules.
mitre_tags	Mapped ATT&CK tactics/techniques where available.
case_refs	Related alert/incident IDs.
enrichment_refs	Threat intel, vulnerability, geo, reputation references.

7.3 Storage Classes

Storage Class	Purpose	Default Retention	Notes
Hot event store	Fast search, detection, investigation, recent dashboards.	30-90 days by package.	Higher tiers may include longer hot retention.
Warm analytics store	Historical analytics, trend reporting, rule testing.	90-365 days by package.	May use cheaper query-optimised storage.
Archive/object storage	Long-term evidence, raw events, reports.	1-7 years by contract/regulation.	Encrypted, access-controlled, legal-hold capable.
Relational case DB	Tenants, users, cases, tasks, configs, reports, billing.	Life of contract plus retention policy.	Strong integrity, backups, audit logs.
Vector/knowledge store	Knowledge base, AI retrieval, detection context.	According to data classification.	Must enforce tenant separation.

Immutable audit store	Security-relevant audit events.	Minimum 1 year; longer for regulated tiers.	Tamper-evident controls required.
-----------------------	---------------------------------	---	-----------------------------------

7.4 Retention and Deletion Requirements

Req ID	Requirement
DATA-001	The system shall support retention policies by tenant, data class, package, jurisdiction, and legal hold status.
DATA-002	The system shall support deletion workflows for tenant offboarding while preserving audit/legal records where contractually or legally required.
DATA-003	The system shall prevent deletion of evidence under active legal hold or active major incident review.
DATA-004	The system shall provide retention preview showing what will be deleted/archived before policy execution.
DATA-005	The system shall log retention policy execution and deletion outcomes.
DATA-006	The system shall support customer export before deletion where contractually allowed.
DATA-007	The system shall support data minimisation and field-level masking for sensitive personal data.
DATA-008	The system shall support customer-specific encryption keys or key namespaces for dedicated/regulated tiers.
DATA-009	The system shall provide data residency metadata and prevent routing data to prohibited regions.
DATA-010	The system shall support disaster recovery restoration without violating tenant isolation or retention policies.

8. Integration and Connector Requirements

8.1 Integration Strategy

The platform shall be integration-first but not integration-dependent. Native case management, alert queue, reporting, AI triage, evidence handling, packages, and SOC workflows must operate with basic syslog, webhook, API, and manual uploads. Vendor-specific connectors increase depth and automation but must be implemented in phases based on market value, customer demand, and technical feasibility.

8.2 Connector Priority Roadmap

Phase	Connector	Target Use	Purpose
P0	Generic API/Webhook Ingest	All customers	Inbound alerts/events/findings/assets; flexible early integration.
P0	Syslog Collector	Firewalls, network devices, Linux, appliances	Broadest generic log source coverage.
P0	Email Notifications	All customers	Customer escalations and service communication.
P1	Microsoft 365	Most enterprises/SMEs	Email, audit, identity, collaboration security context.
P1	Microsoft Entra ID	Identity-first detection	Sign-ins, risky users, groups, privileged identity context; disable/revoke actions.
P1	Microsoft Defender for Endpoint	Endpoint MDR	Endpoint alerts, device isolation, investigation packages.
P1	Teams	Customer approvals and notifications	Collaboration and major incident war-room support.
P1	Jira/ServiceNow	Enterprise service management	Ticket sync and remediation task management.
P2	AWS	Cloud security	CloudTrail, GuardDuty, IAM, EC2, S3 events, response tasks.
P2	Azure	Cloud security	Activity logs, Defender for Cloud, resource inventory.
P2	Google Workspace	Identity/email collaboration	Audit logs and identity/email signals.
P2	Fortinet	Network security	Firewall alerts, syslog, block actions.
P2	Palo Alto	Network/security platform	Firewall, Prisma, Cortex integration potential.
P2	CrowdStrike	Endpoint/XDR	Falcon detections, host context, containment actions.
P2	SentinelOne	Endpoint/XDR	Endpoint detections and response actions.
P2	Sophos	Endpoint/email/network	Endpoint alerts and security context.
P3	Cisco	Network/security	Umbrella, Secure Endpoint, Meraki, firewall/network signals.
P3	Okta	Identity	Identity logs, risk, user/session response.
P3	Cloudflare	Edge/SASE/DNS	DNS, WAF, Zero Trust, blocked traffic.
P3	Zscaler	SASE/SWG	Web access and private access telemetry.
P3	GCP	Cloud security	Cloud audit logs, SCC findings, IAM context.
P3	Vulnerability Scanners	Exposure management	Tenable, Qualys, Rapid7, Defender VM.
P3	Email Security Platforms	Phishing/BEC	Proofpoint, Mimecast, Defender, Barracuda or similar.
P4	Customer Sector APIs	Banks, telecoms, utilities, public sector	Sector-specific telemetry and compliance workflows.

8.3 Standard Connector Requirements

Req ID	Requirement
CONN-001	Every connector shall declare supported data types, auth method, required scopes, polling method, rate limits, retries, pagination, checkpointing, and schema version.
CONN-002	Every connector shall expose health status, last successful run, error count, data volume, event lag, credential expiry, and parser failure rate.
CONN-003	Every connector shall support tenant-scoped credentials and never reuse credentials across tenants.
CONN-004	Every connector shall encrypt stored secrets and restrict secret access to the connector runtime.
CONN-005	Every connector shall support test connection, permissions validation, and guided setup instructions.
CONN-006	Every connector shall support idempotent ingestion and duplicate handling.
CONN-007	Every connector shall record raw payload references, normalized payload status, and mapping version.
CONN-008	Outbound response connectors shall validate authority-to-act before executing actions.
CONN-009	Outbound actions shall capture external action IDs and final status.
CONN-010	Connectors shall fail safely and escalate to analysts if response action status is unknown.

8.4 Connector Contract Template

Contract Element	Required Content
Connector Name	Vendor/product/source and version.
Connector Type	Inbound alert, inbound log, asset inventory, vulnerability, identity, outbound action, ticketing, messaging, threat intelligence.
Authentication	OAuth, API key, certificate, service account, webhook secret, SAML/OIDC, or syslog TLS.
Permissions	Minimum required permissions and optional permissions for response actions.
Data Objects	Alerts, users, endpoints, vulnerabilities, cloud resources, emails, network flows, events, tickets, indicators.
Polling/Streaming	Polling interval, webhook endpoint, syslog listener, batch schedule, backfill support.
Rate Limits	Vendor rate limits and retry/backoff approach.
Normalization Mapping	Canonical fields, parser version, unmapped field handling.
Health Metrics	Last event time, last success, errors, data volume, lag, expiry.
Security	Secret storage, tenant isolation, TLS, IP allowlisting, credential rotation.
Testing	Unit tests, sandbox tests, tenant setup tests, action simulation tests.
Documentation	Customer setup guide, troubleshooting, permissions risk, known limitations.

Confidential - Draft for build planning and expert validation

9. AI SOC, Detection Engineering and SOAR Requirements

9.1 AI Principles

- AI is a copilot and workflow accelerator, not an uncontrolled decision-maker.
- AI outputs must be grounded in platform evidence and distinguish fact from inference.
- AI shall be prevented from cross-tenant leakage by architecture, retrieval scoping, prompt construction, access controls, and logging.
- AI shall never be the sole approver of high-impact containment actions.
- AI recommendations must be reviewable, editable, rejectable, and measurable.
- AI shall support no-AI mode, private-model mode, hosted-model mode, and customer-specific restrictions where commercially required.

9.2 AI Feature Requirements

ID	Feature	Requirement
AI-F-001	Alert summarisation	Generate concise summary of what triggered, affected entity, why it matters, and suggested triage steps.
AI-F-002	Incident narrative	Create an incident story with timeline, likely sequence, affected systems, evidence and remaining unknowns.
AI-F-003	Evidence extraction	Identify key observables, suspicious actions, relevant logs, and missing data for analyst review.
AI-F-004	MITRE mapping assistance	Suggest likely ATT&CK tactics/techniques based on evidence with confidence and rationale.
AI-F-005	Recommended next action	Suggest safe next investigation or response steps based on playbook and authority-to-act.
AI-F-006	Customer update draft	Draft customer-facing technical and executive updates for analyst approval.
AI-F-007	Post-incident report draft	Draft PIR including root cause, impact, response timeline, lessons learned, and recommendations.
AI-F-008	Detection tuning assistant	Summarise false-positive patterns and recommend rule tuning for detection engineer review.
AI-F-009	Threat intel explainer	Explain indicator/campaign relevance in analyst-friendly language.
AI-F-010	Query assistant	Generate search queries from natural language, subject to preview and analyst execution.
AI-F-011	Playbook assistant	Recommend playbook branch or next step, not direct high-risk action execution.
AI-F-012	Executive summary	Create board-ready monthly cyber posture summary using service metrics and incident context.

9.3 AI Control Requirements

Req ID	Requirement
AI-C-001	AI retrieval shall be tenant-scoped and permission-filtered.
AI-C-002	AI context packages shall include only necessary evidence and be logged.
AI-C-003	AI responses shall cite source records or state when confidence is low.
AI-C-004	AI shall not invent evidence, dates, indicators, or actions.
AI-C-005	AI-generated customer communications shall require human approval before external delivery.
AI-C-006	AI-generated response actions shall require playbook approval workflow.
AI-C-007	The system shall test AI features for prompt injection using malicious log/event content.
AI-C-008	The system shall provide model/provider configuration and data routing restrictions by tenant.
AI-C-009	The system shall maintain feedback labels: useful, incorrect, unsafe, hallucinated, insufficient evidence, accepted, edited.
AI-C-010	The system shall periodically evaluate AI outputs against analyst-reviewed ground truth.

9.4 Detection Lifecycle

Stage	Required Controls	Outputs
Intake	Threat intel, customer request, incident lesson learned, coverage gap, regulatory need.	Detection request ticket.
Design	Define logic, data sources, fields, severity, false positives, ATT&CK mapping, playbook.	Detection design record.
Author	Create Sigma/native rule and tests.	Rule draft and test data.
Peer Review	Detection engineer and SOC analyst review for logic and operational value.	Review approval/rework.
Test	Run against sample and historical data where allowed.	Test results and expected alert examples.
Pilot	Deploy to limited tenant/group or shadow mode.	Pilot findings and FP rate.
Production	Deploy with monitoring and rollback plan.	Active rule version.
Tune	Adjust thresholds, exclusions, context, severity.	Tuning record.
Retire	Remove obsolete noisy/unsafe rules.	Retirement reason and replacement mapping.

9.5 SOAR Action Risk Classes

Class	Examples	Approval Requirement
Class 0 - Informational	Enrich indicator, retrieve asset profile, create internal note.	No approval; system logged.
Class 1 - Low Impact	Create ticket, notify internal analyst, add watchlist item.	Policy-based approval optional.
Class 2 - Medium Impact	Customer notification, request password reset, mark email for review.	Analyst approval required unless package pre-authorises.

Best-in-Class Cyber Security Operations Platform - Full SRS

Class 3 - High Impact	Disable user, revoke sessions, isolate endpoint, block business domain/IP.	Customer or authorised senior approval required unless explicit pre-authorised containment.
Class 4 - Business Critical	Network-wide block, mass quarantine, cloud account lockdown, production service disruption.	Incident commander + customer authority required; no full autonomous execution in MVP/V1.

10. Incident Response, SOC Operations and Service Management Requirements

10.1 SOC Tiers and Responsibilities

Role	Responsibilities	Platform Needs
Tier 1 Analyst	Queue monitoring, initial triage, enrichment review, false-positive closure, escalation, customer notification drafts.	Fast queues, playbooks, AI summary, closure templates, SLA timers.
Tier 2 Analyst	Deep investigation, timeline building, containment recommendations, customer technical coordination.	Entity graph, advanced search, notebooks, response approval, evidence handling.
Tier 3 / IR Lead	Major incidents, threat hunting, response strategy, forensic readiness, PIR leadership.	War room, chain-of-custody, advanced queries, high-risk actions, PIR templates.
Detection Engineer	Rule creation/tuning, coverage, test lab, content QA.	Detection workspace, rule tests, FP feedback, coverage dashboard.
SOC Manager	Shift control, QA, queue health, analyst performance, service review.	Ops dashboard, utilisation, QA sampling, escalation metrics.
Customer Success Manager	Onboarding, service reviews, customer satisfaction, renewal risk.	Customer health, reports, open tasks, contract entitlements.
Compliance Manager	Evidence packs, audit response, regulatory mapping.	Control mapping, evidence workflows, export approvals.

10.2 Severity Model

Severity	Definition	Examples	Response Expectations
P1 Critical	Confirmed or highly likely active compromise causing or likely to cause material business impact.	Ransomware activity, privileged account takeover, active exfiltration, crown-jewel compromise.	Immediate triage, senior escalation, customer emergency notification, war-room mode.
P2 High	Credible threat requiring urgent investigation and potential containment.	Malware on critical host, impossible travel privileged login, EDR high confidence detection.	Priority queue, Tier 2 escalation, customer notification per SLA.
P3 Medium	Suspicious activity requiring investigation but no immediate material impact evident.	Phishing clicked, unusual sign-in, suspicious PowerShell, exposed vulnerable asset.	Standard triage and customer task as needed.
P4 Low	Low confidence or policy/security hygiene finding.	Blocked commodity malware, failed brute-force attempts, benign policy alert.	Triage, tune/suppress if noisy, monthly reporting if applicable.
P5 Informational	Awareness, health, or advisory item.	Connector health, threat advisory, control observation.	Notification/reporting only.

10.3 Minimum SOC Process Requirements

Req ID	Requirement
OPS-001	Every shift shall start with queue review, major incident review, open customer approval review, integration health review, and handover notes.
OPS-002	Every P1/P2 incident shall have named incident owner, severity rationale, affected entities, next action, and customer communication status.
OPS-003	Every incident shall maintain a timeline of system events, analyst actions, customer actions, AI assistance, and playbook actions.
OPS-004	Every closure shall include disposition, root cause if known, impact, actions taken, customer tasks, and detection feedback.
OPS-005	Every false positive shall capture reason category and rule/data-source feedback.
OPS-006	Every major incident shall produce a post-incident report and lessons-learned action list.
OPS-007	Every customer shall have onboarding baseline, escalation matrix, critical asset list, and authority-to-act settings before production monitoring.
OPS-008	Every month, each managed customer shall receive a service report unless package excludes it.
OPS-009	SOC management shall sample and QA closed cases for quality and consistency.
OPS-010	The platform shall support audit of analyst workload, queue ageing, SLA breaches, and customer waiting states.

10.4 Core Playbook Catalogue

ID	Playbook	Required Content
PB-001	Phishing / Suspicious Email	Email headers, URL reputation, sender domain, user clicked, attachment hash, mailbox search, quarantine request, user guidance.
PB-002	Business Email Compromise	Mailbox rules, sign-in anomalies, MFA status, forwarding rules, OAuth consent, financial system exposure, urgent containment.
PB-003	Ransomware Indicators	EDR alerts, mass file changes, suspicious encryption tools, lateral movement, isolate endpoint, backup/service impact checks.
PB-004	Endpoint Malware	EDR detection, hash/process tree, host isolation, scan, user context, related hosts, closure after clean state.
PB-005	Privileged Account Compromise	Risky sign-in, impossible travel, privilege change, session revocation, password reset, PAM review.
PB-006	Cloud IAM Abuse	New access keys, privilege escalation, unusual API calls, cloud trail review, key disablement approval.
PB-007	Data Exfiltration	Large outbound transfer, unusual destination, DLP alerts, affected data class, containment and legal/privacy escalation.
PB-008	Vulnerability Exploitation	Exploit intel, vulnerable assets, WAF/firewall events, patch/workaround tasks, customer risk acceptance.
PB-009	Firewall / Network Intrusion	IDS/IPS events, blocked/allowed flow, asset criticality, threat intel, block rule request.
PB-010	Suspicious PowerShell / Script Execution	Command line, parent process, user context, host criticality, known admin tools vs attacker tradecraft.
PB-011	New Persistence Mechanism	Scheduled task, service creation, startup folder, registry run key, cloud persistence, containment plan.
PB-012	DDoS / Service Availability	Traffic anomaly, provider telemetry, customer business impact, mitigation provider escalation.
PB-013	Insider Risk Indicator	Unusual access, data movement, privileged action, HR/legal handling, confidentiality controls.
PB-014	Third-Party/Supply Chain Alert	Vendor compromise advisory, exposed integration, indicators, affected customers, notification and mitigation.
PB-015	Connector Health Failure	Critical data-source outage, customer alert, remediation task, SLA impact, compensating monitoring.

11. Compliance, Regulatory Evidence and Governance Requirements

11.1 Compliance Product Design

Compliance shall be implemented as configurable mappings and evidence workflows rather than hardcoded country-specific logic. Country-specific packs such as Ghana sovereign SOC, Nigerian regulated enterprise, UK financial services, EU-style data protection, or public-sector CII shall be deployable as templates containing control mappings, reporting calendars, evidence expectations, report templates, and data-residency rules.

11.2 Governance Requirements

Req ID	Requirement
GOV-001	The platform shall maintain an immutable audit trail for all administrative, analyst, API, automation, AI, export, and security-sensitive actions.
GOV-002	The platform shall support internal governance dashboards for risk, audit findings, control exceptions, data retention, access review, and operational incidents.
GOV-003	The platform shall support customer-facing assurance reports summarising security controls, availability, incident handling process, and evidence handling.
GOV-004	The platform shall support mapping of service obligations to customer contracts and packages.
GOV-005	The platform shall support documented shared responsibility model per deployment/package.
GOV-006	The platform shall track policy acknowledgements by customers and internal users.
GOV-007	The platform shall support exportable evidence for external audits and regulatory reviews.
GOV-008	The platform shall support escalation and reporting workflows for suspected data breach, security incident, service outage, or SLA breach.
GOV-009	The platform shall support separation of duties for rule deployment, privileged admin, billing override, data export, and high-risk response actions.
GOV-010	The platform shall support regular access review and certification.

11.3 Evidence Pack Requirements

Evidence Pack Type	Included Evidence
Monthly Service Review	Open/closed incidents, SLA performance, integration health, false positives, top recurring detections, recommendations, open customer tasks.
Board Cyber Posture	Risk trends, major incidents, attack themes, control gaps, exposure summary, business impact, management actions.
Regulatory/CII Evidence	Monitoring coverage, incident handling timeline, data-source health, response actions, escalation logs, control mapping.
Post-Incident Report	Executive summary, technical timeline, affected assets/users, root cause, impact, containment/eradication/recovery actions, lessons learned.
Audit Evidence Bundle	Access logs, detection records, incident samples, evidence chain, reports, approvals, playbook execution logs.
Customer Onboarding Evidence	Scope, contacts, assets, authority-to-act, connectors, baseline tuning, test alerts, go-live sign-off.

12. Non-Functional Requirements

Availability and Resilience

Req ID	Requirement
NFR-AV-001	The system shall be designed for high availability with redundant critical services and graceful degradation.
NFR-AV-002	The system shall define SLOs for API availability, ingestion availability, alert processing, report generation, and UI access.
NFR-AV-003	The system shall provide disaster recovery with defined RPO/RTO per deployment tier.
NFR-AV-004	The system shall support queue buffering during downstream outages.
NFR-AV-005	The system shall provide customer-visible status for integration and platform outages.

Performance and Scalability

Req ID	Requirement
NFR-PERF-001	The system shall process high-volume event streams with horizontal scaling of ingestion, parsing, detection, enrichment, and storage.
NFR-PERF-002	The analyst UI shall load priority queues within acceptable interactive response times under normal load.
NFR-PERF-003	Search shall support time-bounded queries over hot data with predictable performance.
NFR-PERF-004	Report generation shall be asynchronous for large reports.
NFR-PERF-005	The platform shall support capacity forecasting and usage trends by tenant.

Security

Req ID	Requirement
NFR-SEC-001	The platform shall be secure-by-design with threat modelling for major features.
NFR-SEC-002	All network communications shall use strong TLS.
NFR-SEC-003	Data at rest shall be encrypted using managed keys or customer/dedicated keys where applicable.
NFR-SEC-004	Secrets shall be stored in a secrets manager and never in code or logs.
NFR-SEC-005	The platform shall support regular vulnerability scanning, dependency scanning, SAST, DAST, IaC scanning, and penetration testing.
NFR-SEC-006	The platform itself shall be monitored by its own security controls and alerting.
NFR-SEC-007	Production access shall be controlled through least privilege, MFA, logging, and approval for privileged actions.
NFR-SEC-008	Every data export shall be permission checked and logged.
NFR-SEC-009	The system shall prevent cross-tenant leakage at application, database, cache, search, object store, report, AI, and logging layers.
NFR-SEC-010	The system shall provide tamper-evident audit logging for sensitive actions.

Privacy and Data Protection

Req ID	Requirement
NFR-PRI-001	The platform shall support data minimisation, masking, pseudonymisation, retention, deletion, export, and legal hold policies.
NFR-PRI-002	The platform shall classify sensitive personal data and restrict access by role.
NFR-PRI-003	The platform shall prevent sending prohibited data to external AI or third-party services based on tenant policy.
NFR-PRI-004	The platform shall support data residency restrictions.
NFR-PRI-005	The platform shall maintain data-processing logs and evidence for privacy governance.

Usability and Accessibility

Req ID	Requirement
NFR-UX-001	The analyst workbench shall minimise clicks during triage and investigation.
NFR-UX-002	High-risk actions shall use clear confirmation prompts explaining impact.
NFR-UX-003	Dashboards shall be understandable by non-technical executives.
NFR-UX-004	The UI shall support keyboard navigation, sufficient contrast, visible focus states, and clear labels.
NFR-UX-005	The platform shall provide guided onboarding and connector setup instructions.

Maintainability and Extensibility

Req ID	Requirement
NFR-MA-001	The platform shall be modular with documented APIs and internal contracts.
NFR-MA-002	Connectors, parsers, detections, playbooks, report templates, and compliance packs shall be versioned content.
NFR-MA-003	The codebase shall include automated tests for critical workflows and permission boundaries.
NFR-MA-004	The platform shall support feature flags and controlled rollout.
NFR-MA-005	The platform shall provide developer documentation sufficient for Developers and human engineers.

13. Security, Privacy and Multi-Tenancy Requirements

13.1 Multi-Tenancy Security Controls

Req ID	Requirement
MT-001	Every request shall be tenant-scoped through verified user/session/API context.
MT-002	Every query shall enforce tenant predicates and permission filters at service and data-access layers.
MT-003	Object storage paths and keys shall include tenant isolation boundaries.
MT-004	Search indexes shall be tenant-partitioned or permission-filtered with tests for cross-tenant leakage.
MT-005	Caches shall be scoped by tenant and user permissions.
MT-006	AI retrieval shall be tenant-scoped and shall not retrieve or embed cross-tenant content.
MT-007	Report templates and scheduled reports shall validate tenant scope at generation time and delivery time.
MT-008	Platform administrators shall require explicit tenant impersonation workflow with reason and audit log if support access is needed.
MT-009	Dedicated tenants shall support separate keys, data stores, or namespaces depending on package.
MT-010	Automated tests shall include negative tests attempting cross-tenant access for all major APIs.

13.2 Secure SDLC Requirements

Req ID	Requirement
SSDLC-001	All code shall be peer-reviewed or senior-engineer-reviewed, including AI-generated code.
SSDLC-002	CI/CD shall run unit tests, integration tests, linting, type checks, SAST, dependency scanning, container scanning, and IaC scanning.
SSDLC-003	Production deployments shall require controlled release process with rollback plan.
SSDLC-004	Secrets shall be injected at runtime and never committed.
SSDLC-005	Threat modelling shall be performed for ingestion, identity, multi-tenancy, SOAR actions, AI, report exports, and connectors.
SSDLC-006	Security defects shall be tracked with severity, SLA, owner, and verification.
SSDLC-007	The platform shall maintain SBOM and dependency update process.
SSDLC-008	Penetration testing shall be performed before enterprise launch and after major architectural changes.
SSDLC-009	All privileged admin tools shall be protected from accidental or unauthorised use.
SSDLC-010	Customer-facing security documentation shall be maintained.

14. User Journeys and Use Cases

UC-001 - Onboard New SOCaaS Customer

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Tenant admin and onboarding engineer configure customer profile, package, contacts, authority-to-act, critical assets, connectors, test alerts, reports, and go-live sign-off.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-002 - Connect Microsoft 365/Entra ID

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Tenant admin grants least-privilege permissions; platform validates scopes, tests ingestion, maps identity events, and displays health.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-003 - Triage New High-Severity Alert

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Tier 1 analyst receives enriched alert, reviews AI summary, checks entity graph, runs playbook steps, escalates or closes with reason.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-004 - Convert Alert to Incident

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Analyst groups related alerts, creates incident, assigns severity, starts SLA, adds affected assets/users, and notifies customer if required.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-005 - Customer Approves Containment

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	System sends approval request with evidence and impact; authorised customer approves; SOAR executes action and logs result.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-006 - Run Threat Hunt

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Tier 3 analyst selects hunt hypothesis, queries telemetry, creates findings, converts confirmed findings to cases, and updates detections.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-007 - Detection Engineer Tunes Noisy Rule

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Engineer reviews false-positive feedback, tests change, deploys tuned rule to pilot tenants, then promotes to production.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-008 - Generate Monthly Service Report

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Customer success manager reviews report draft, validates metrics, adds commentary, approves, and distributes securely.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-009 - Generate Regulatory Evidence Pack

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Compliance manager selects framework, time period, incidents, controls, evidence, and exports regulator/auditor-ready pack.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-010 - Handle Major Ransomware Incident

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Platform opens war-room, escalates P1, links alerts, runs containment approvals, captures timeline, supports PIR generation.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-011 - MSSP Creates Downstream Tenant

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Partner admin creates client tenant, applies package template, white-label settings, and assigns partner analyst team.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-012 - Offboard Customer

Item	Detail
------	--------

Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Admin freezes new ingestion, exports agreed data, applies retention/legal hold, revokes connectors, archives tenant, and records closure certificate.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-013 - AI Drafts Incident Summary

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Analyst requests summary; AI retrieves scoped evidence, drafts facts/inferences/action items, analyst edits and approves.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-014 - Connector Failure Escalation

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Critical connector fails; platform alerts SOC and customer, creates task, records SLA impact, and shows monitoring gap.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

UC-015 - Board Views Cyber Posture

Item	Detail
Primary Actor	Defined by use case: customer admin, SOC analyst, detection engineer, compliance manager, MSSP partner or executive.
Goal	Executive user views risk trends, major incidents, remediation status, service performance, and recommendations without raw technical detail.
Preconditions	Actor is authenticated; tenant/package permissions exist; required data sources and policies are configured as applicable.
Main Flow	User starts workflow, platform validates permissions, presents relevant data, captures actions, updates records, logs audit trail, and triggers notifications or next tasks.
Alternate / Exception Flow	Missing permissions, connector unavailable, data incomplete, approval not received, action fails, or SLA breach occurs; system escalates and preserves state.
Acceptance Criteria	Workflow can be completed end-to-end in test environment; audit log captures all material actions; tenant isolation verified; error handling tested.

15. Service Packages, Entitlements and Commercial Guardrails

15.1 Package Model

Package	Target	Included Capabilities	Commercial Guardrail
Essential Monitor	SME/small agency	Core monitoring, limited assets/logs, basic reports, email escalation, limited connectors.	No high-risk automated response; clear log/asset limits.
Standard SOCaaS	Mid-market/MDA	24/7 monitoring, core identity/endpoint/firewall logs, monthly reports, ticketing, basic playbooks.	Overage billing for log volume and custom integrations.
Advanced MDR	Regulated entity	Expanded logs, EDR/XDR, threat intelligence, response playbooks, quarterly review, compliance pack.	Authority-to-act policy mandatory; senior analyst escalation.
Critical Infrastructure	Bank/telco/utility/major GovTech	Priority queue, custom detections, senior escalation, IR retainer, board dashboard, longer retention.	Minimum annual commitment; stricter onboarding readiness.
Enterprise Private SOC	Large enterprise	Dedicated instance or dedicated data plane, co-managed SOC, custom integrations.	Professional services SOW and support model.
Sovereign SOC	Public-sector/CII national or regional programme	Country-hosted, local compliance pack, sector dashboards, anchor tenant framework.	Formal legal/regulatory positioning; no implied national mandate without authority.
MSSP White-Label	Managed service provider	Partner tenant hierarchy, branding, downstream customer management, partner analyst access.	Partner contract, support obligations, downstream data controls.

15.2 Entitlement Requirements

Req ID	Requirement
ENT-001	The platform shall enforce package limits for assets, log volume, connectors, users, retention, reports, AI usage, playbook actions, and response hours.
ENT-002	The platform shall provide customer-visible usage against entitlement.
ENT-003	The platform shall alert internal commercial team before customers exceed limits.
ENT-004	The platform shall require approval for custom scope not included in package.
ENT-005	The platform shall prevent analysts from unknowingly delivering out-of-scope custom work without capturing billable task or exception.
ENT-006	The platform shall support package upgrade/downgrade workflows.
ENT-007	The platform shall preserve historic entitlements for audit and billing dispute resolution.
ENT-008	The platform shall support framework/anchor customers without allowing unlimited customisation to destroy margin.

16. MVP, V1, V2 and Enterprise Roadmap

Phase	Indicative Timing	Scope
Phase 0 - Design Mobilisation	0-8 weeks	Final SRS validation, architecture decisions, threat model, UX prototypes, connector priority, vendor choices, delivery governance, founding cyber expert review.
Phase 1 - MVP SOC Core	Months 1-4	Multi-tenant core, auth/RBAC, customer/admin portals, analyst queue, basic alert API/webhook/syslog ingestion, case management, notes/tasks/evidence, basic reports, email/Teams notification, audit logs.
Phase 2 - SOCaaS V1	Months 4-8	Microsoft 365/Entra/Defender connectors, integration health, package entitlements, SLA timers, AI alert summaries, core playbooks, monthly service reports, onboarding workflow.
Phase 3 - MDR/XDR V1	Months 8-14	EDR/firewall/cloud connectors, correlation, entity graph, threat intel, detection-as-code, SOAR approvals, vulnerability context, advanced incident reports.
Phase 4 - Enterprise/Sovereign	Months 14-24	Dedicated tenant/data plane, compliance packs, white-label MSSP, extended retention, advanced dashboards, country deployment templates, sector playbooks.
Phase 5 - Agentic SOC Optimisation	Months 18-30	AI investigation workflows, safe action recommendations, detection tuning assistant, threat hunting copilot, continuous coverage scoring, advanced automation governance.

16.1 MVP Must-Have Scope

- Tenant/account management
- RBAC/MFA basic implementation
- Customer/admin/analyst portals
- Alert ingestion API, webhook, and syslog collector
- Normalized alert model
- Case management lifecycle
- Tasks, notes, evidence attachments
- SLA timers
- Audit logging
- Basic reporting
- Email/Teams notifications
- Connector health basics
- Package entitlements basics
- Secure deployment pipeline
- Initial Microsoft connector design
- AI summary prototype with strict controls

16.2 Deferred from MVP

- Full custom SIEM replacement
- All vendor connectors
- Advanced anomaly detection
- Fully autonomous response
- Mobile apps
- Regulator portal
- Advanced white-label marketplace
- Full billing integration
- Complex OT/industrial monitoring
- External cyber insurance integration

17. Implementation Plan and Build Governance

17.1 Delivery Team

Role	Minimum Responsibility
Product Owner	Owns vision, scope, commercial packages, prioritisation, stakeholder decisions.
Cybersecurity Architect	Defines SOC/XDR architecture, data sources, detection design, response safety, control mapping.
Solution Architect	Defines system architecture, services, APIs, data model, deployment model, scalability.
Senior Full-Stack Engineer	Reviews AI-generated code, sets engineering standards, owns critical modules.
DevSecOps/Cloud Engineer	Builds secure CI/CD, infrastructure, secrets, monitoring, environments.
SOC Operations Lead	Defines shift process, severity, escalation, runbooks, QA and service management.
Detection Engineer	Builds detection library, tuning workflow, MITRE coverage, rule tests.
Data Engineer	Builds ingestion, normalization, storage, parser framework.
UX Designer	Designs analyst-efficient workflows and customer dashboards.
QA/Security Tester	Executes functional, integration, security, permission, performance and regression tests.
Legal/Compliance Advisor	Reviews privacy, data residency, contracts, authority-to-act, liability, reporting obligations.

17.2 Coding Governance

Req ID	Requirement
AGENT-001	Developers may generate code, tests, migration scripts, documentation, connector scaffolds, and UI components and ensure to review properly including architecture and review.
AGENT-002	Every generated module shall have tests and code review before merge.
AGENT-003	Security-sensitive modules such as auth, RBAC, tenant isolation, SOAR actions, AI context retrieval, encryption, and secrets shall require additional self review
AGENT-004	Prompts and task tickets shall reference specific SRS requirement IDs.
AGENT-005	All coding output shall be scanned for insecure patterns, dependency risks, licensing issues, and data leakage.
AGENT-006	The product backlog shall maintain traceability between SRS requirements, code modules, tests, and release acceptance.

17.3 Environment Strategy

Environment	Purpose	Controls
Local Development	Developer and AI coding agent implementation.	Synthetic data only, no secrets, lint/test hooks.
Shared Dev	Integrated feature testing.	Seed data, mock connectors, ephemeral credentials.
Security Test	Threat modelling validation, SAST/DAST, tenancy tests, abuse tests.	Restricted access, test tenants, exploit simulations.
UAT / Pilot	Customer pilot and SOC process validation.	Limited customer data, sign-off, strict audit.
Production	Live SOC service.	MFA, least privilege, deployment approval, monitoring, backups.
DR	Disaster recovery environment.	Tested restoration, controlled failover, periodic drills.

18. Testing, Validation and Acceptance Criteria

18.1 Test Categories

Test Category	Coverage
Unit Tests	Business logic, parsers, rule evaluation, API validation, utility functions.
Integration Tests	Connectors, queues, storage, workflow engine, notification systems, AI service calls.
End-to-End Tests	Customer onboarding, alert ingestion to case, incident lifecycle, report generation, approval workflow.
Security Tests	Auth, RBAC/ABAC, tenant isolation, injection, SSRF, secrets, export, audit logs, SOAR action safety.
Performance Tests	Ingestion throughput, queue latency, search, dashboard load, report generation, API rate limits.
Resilience Tests	Connector outage, queue backlog, storage outage, AI service unavailable, partial playbook failure.
Data Quality Tests	Parser mapping, schema drift, unmapped fields, entity resolution, duplicate handling.
AI Evaluation	Grounding, hallucination, unsafe action recommendation, prompt injection, tenant leakage, factual accuracy.
SOC UAT	Analyst triage, escalation, closure, reporting, handover, major incident war-room.
Customer UAT	Portal access, report understanding, approval workflow, tasks, executive dashboard.
Compliance Tests	Evidence pack generation, access review, audit trail completeness, retention policy.
Regression Tests	No breakage to prior features after new connector/rule/playbook release.

18.2 Acceptance Gates

Gate	Acceptance Standard
Gate 1 - Architecture	Architecture approved by cyber architect, solution architect, senior engineer, and product owner.
Gate 2 - Secure Foundation	Auth, RBAC, tenant isolation, audit logs, secrets, encryption, CI/CD and baseline infrastructure pass security tests.
Gate 3 - MVP Functional	Core tenant, ingestion, case, task, evidence, reporting, and notifications complete with tests.
Gate 4 - SOC Operational Pilot	SOC analysts can operate sample alerts through triage, escalation, customer notification, and closure.
Gate 5 - Connector Pilot	Priority connectors ingest data reliably and expose health/error/volume metrics.
Gate 6 - AI Safety	AI features pass grounding, permission, prompt injection, and human-approval tests.
Gate 7 - Commercial Pilot	Package entitlements, usage metering, service reports, and customer onboarding run end-to-end.
Gate 8 - Production Readiness	Monitoring, backups, DR plan, incident management, security testing, and runbooks are approved.

18.3 Definition of Done

- Requirement ID mapped to user story and test cases.
- Code reviewed by senior engineer.
- Security-sensitive code reviewed by security lead.
- Unit/integration tests passed.
- Tenant isolation test passed where applicable.
- Audit logging implemented where applicable.
- Documentation updated.
- Operational runbook updated if SOC process affected.
- Feature flag and rollback plan created for risky changes.
- Product owner accepts feature in UAT.

19. Requirements Traceability Matrix

The build backlog shall maintain a live traceability matrix mapping SRS requirement IDs to epics, user stories, code repositories, API endpoints, database tables, test cases, release versions, and operational runbooks. The sample below establishes the required structure.

Req ID	Requirement Area	Epic	Story Mapping	Test Mapping	Phase
TEN-001	Tenant, Organisation and Account Management	Epic-TEN	User stories to be decomposed	Unit + integration + UAT	MVP/V1
IAM-001	Identity, Access Management and Session Security	Epic-IAM	User stories to be decomposed	Unit + integration + UAT	MVP/V1
UI-001	User Experience, Portals and Dashboards	Epic-UI	User stories to be decomposed	Unit + integration + UAT	MVP/V1
ING-001	Data Ingestion, Collection and Connector Framework	Epic-ING	User stories to be decomposed	Unit + integration + UAT	MVP/V1
NORM-001	Normalization, Enrichment and Entity Resolution	Epic-NORM	User stories to be decomposed	Unit + integration + UAT	MVP/V1
DET-001	Detection Engineering and Rules Management	Epic-DET	User stories to be decomposed	Unit + integration + UAT	MVP/V1
COR-001	Alert Correlation, Risk Scoring and Prioritisation	Epic-COR	User stories to be decomposed	Unit + integration + UAT	MVP/V1
CASE-001	Alert, Incident and Case Management	Epic-CASE	User stories to be decomposed	Unit + integration + UAT	MVP/V1
INV-001	Investigation Workbench, Timeline and Entity Graph	Epic-INV	User stories to be decomposed	Unit + integration + UAT	MVP/V1
TI-001	Threat Intelligence Management	Epic-TI	User stories to be decomposed	Unit + integration + UAT	MVP/V1
DATA-001	Cross-cutting control	Epic-Control	Cross-cutting story	Security + audit tests	MVP
CONN-001	Cross-cutting control	Epic-Control	Cross-cutting story	Security + audit tests	MVP
AI-C-001	Cross-cutting control	Epic-Control	Cross-cutting story	Security + audit tests	MVP
OPS-001	Cross-cutting control	Epic-Control	Cross-cutting story	Security + audit tests	MVP
GOV-001	Cross-cutting control	Epic-Control	Cross-cutting story	Security + audit tests	MVP
NFR-SEC-001	Cross-cutting control	Epic-Control	Cross-cutting story	Security + audit tests	MVP
MT-001	Cross-cutting control	Epic-Control	Cross-cutting story	Security + audit tests	MVP
AGENT-001	Cross-cutting control	Epic-Control	Cross-cutting story	Security + audit tests	MVP

20. Appendices

Appendix A - Detection Use Case Library Seed

ID	Detection Use Case	Domain	MITRE Tactic	Required Sources	Default Priority
DUC-001	Impossible travel for privileged user	Identity	Credential Access	Entra/Okta/Google Workspace	P2
DUC-002	MFA disabled for privileged account	Identity	Defense Evasion	Entra/Okta	P1
DUC-003	New inbox forwarding rule to external domain	Email/Identity	Collection/Exfiltration	M365/Google Workspace	P2
DUC-004	Suspicious OAuth application consent	Identity/Cloud	Persistence	M365/Entra/Google	P2
DUC-005	Endpoint ransomware behaviour	Endpoint	Impact	Defender/CrowdStrike/SentinelOne/Sophos	P1
DUC-006	Mass file rename/encryption pattern	Endpoint/File	Impact	EDR/Windows logs	P1
DUC-007	Suspicious PowerShell encoded command	Endpoint	Execution	Windows/EDR	P2
DUC-008	Credential dumping tool detected	Endpoint	Credential Access	EDR	P1
DUC-009	Lateral movement using remote services	Endpoint/Network	Lateral Movement	EDR/Windows/Network	P1
DUC-010	New local admin account created	Endpoint/Identity	Persistence/Privilege Escalation	Windows/EDR	P2
DUC-011	Cloud access key created by unusual principal	Cloud	Persistence	AWS/Azure/GCP	P2
DUC-012	Cloud security group opened to world	Cloud/Network	Initial Access	AWS/Azure/GCP	P2
DUC-013	S3/blob bucket made public	Cloud	Exfiltration/Exposure	AWS/Azure/GCP	P2
DUC-014	GuardDuty/SCC/Defender high finding	Cloud	Various	AWS/Azure/GCP	P2
DUC-015	Firewall allowed connection to known C2	Network	Command and Control	Firewall/Threat Intel	P1
DUC-016	Multiple failed VPN logins followed by success	Identity/Network	Initial Access	VPN/Firewall/IdP	P2
DUC-017	Large outbound transfer to new country	Network/Cloud	Exfiltration	Firewall/Proxy/Cloud	P1
DUC-018	DNS tunneling indicator	Network	Command and Control	DNS/Cloudflare/Zscaler	P2
DUC-019	Critical vulnerability exploited on internet-facing asset	Vulnerability/Network	Initial Access	Vuln scanner/WAF/Firewall	P1
DUC-020	Web shell indicator	Endpoint/Web	Persistence	EDR/Web logs	P1
DUC-021	Suspicious email attachment opened	Email/Endpoint	Initial Access	Email security/EDR	P2
DUC-022	User reports phishing and similar messages found	Email	Initial Access	Email platform	P3
DUC-023	Data access spike by insider	Identity/Data	Collection	IdP/DLP/Business API	P2
DUC-024	Admin login from anonymous proxy	Identity	Defense Evasion	IdP/Threat Intel	P2
DUC-025	EDR agent disabled on critical host	Endpoint	Defense Evasion	EDR	P1
DUC-026	Backup deletion or tampering	Endpoint/Cloud	Impact	Cloud/Endpoint/Backup	P1
DUC-027	Suspicious scheduled task creation	Endpoint	Persistence	Windows/EDR	P2
DUC-028	New malicious domain matches tenant email campaign	Threat Intel/Email	Initial Access	TI/Email	P2
DUC-029	OT-adjacent network anomaly	Network/OT	Discovery/Lateral Movement	Network/Syslog/Custom	P1
DUC-030	Connector outage for critical data source	Platform Health	Monitoring Gap	Platform	P2

Appendix B - Initial Product Backlog Seed

Epic ID	Epic	Story ID	User Story	Acceptance Criteria Summary	Phase
EPIC-01	Foundation, Auth and Tenancy	EPIC-01-US-01	As an authorised user, I need foundation, auth and tenancy capability 1 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-01	Foundation, Auth and Tenancy	EPIC-01-US-02	As an authorised user, I need foundation, auth and tenancy capability 2 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-01	Foundation, Auth and Tenancy	EPIC-01-US-03	As an authorised user, I need foundation, auth and tenancy capability 3 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-01	Foundation, Auth and Tenancy	EPIC-01-US-04	As an authorised user, I need foundation, auth and tenancy capability 4 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-01	Foundation, Auth and Tenancy	EPIC-01-US-05	As an authorised user, I need foundation, auth and tenancy capability 5 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-02	Ingestion and Normalisation	EPIC-02-US-01	As an authorised user, I need ingestion and normalisation capability 1 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-02	Ingestion and Normalisation	EPIC-02-US-02	As an authorised user, I need ingestion and normalisation capability 2 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-02	Ingestion and Normalisation	EPIC-02-US-03	As an authorised user, I need ingestion and normalisation capability 3 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-02	Ingestion and Normalisation	EPIC-02-US-04	As an authorised user, I need ingestion and normalisation capability 4 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-02	Ingestion and Normalisation	EPIC-02-US-05	As an authorised user, I need ingestion and normalisation capability 5 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-03	Alert and Case Management	EPIC-03-US-01	As an authorised user, I need alert and case management capability 1 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-03	Alert and Case Management	EPIC-03-US-02	As an authorised user, I need alert and case management capability 2 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-03	Alert and Case Management	EPIC-03-US-03	As an authorised user, I need alert and case management capability 3 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	MVP
EPIC-03	Alert and Case Management	EPIC-03-US-04	As an authorised user, I need alert and case management capability 4 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-03	Alert and Case Management	EPIC-03-US-05	As an authorised user, I need alert and case management capability 5 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2

Best-in-Class Cyber Security Operations Platform - Full SRS

[illegible]

Confidential - Draft for build planning and expert validation

Best-in-Class Cyber Security Operations Platform - Full SRS

			capability 3 so that the platform can deliver secure SOC operations.	requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	
EPIC-10	Compliance and Governance	EPIC-10-US-04	As an authorised user, I need compliance and governance capability 4 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-10	Compliance and Governance	EPIC-10-US-05	As an authorised user, I need compliance and governance capability 5 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-11	Packages and Billing	EPIC-11-US-01	As an authorised user, I need packages and billing capability 1 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-11	Packages and Billing	EPIC-11-US-02	As an authorised user, I need packages and billing capability 2 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-11	Packages and Billing	EPIC-11-US-03	As an authorised user, I need packages and billing capability 3 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-11	Packages and Billing	EPIC-11-US-04	As an authorised user, I need packages and billing capability 4 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-11	Packages and Billing	EPIC-11-US-05	As an authorised user, I need packages and billing capability 5 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-12	MSSP/Sovereign/Enterprise Deployment	EPIC-12-US-01	As an authorised user, I need mssp/sovereign/enterprise deployment capability 1 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-12	MSSP/Sovereign/Enterprise Deployment	EPIC-12-US-02	As an authorised user, I need mssp/sovereign/enterprise deployment capability 2 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-12	MSSP/Sovereign/Enterprise Deployment	EPIC-12-US-03	As an authorised user, I need mssp/sovereign/enterprise deployment capability 3 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-12	MSSP/Sovereign/Enterprise Deployment	EPIC-12-US-04	As an authorised user, I need mssp/sovereign/enterprise deployment capability 4 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2
EPIC-12	MSSP/Sovereign/Enterprise Deployment	EPIC-12-US-05	As an authorised user, I need mssp/sovereign/enterprise deployment capability 5 so that the platform can deliver secure SOC operations.	Acceptance criteria shall reference SRS requirement IDs, permissions, audit logging, tests, and tenant isolation where applicable.	V1/V2

Appendix C - Data Dictionary Seed

Entity	Field	Type	Description	Requirement	Owner
Tenant	id	UUID/string	Unique record identifier	Required	System
Tenant	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Tenant	created_at/updated_at	timestamp	Audit timestamps	Required	System
User	id	UUID/string	Unique record identifier	Required	System
User	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
User	created_at/updated_at	timestamp	Audit timestamps	Required	System
Role/Permission	id	UUID/string	Unique record identifier	Required	System
Role/Permission	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Role/Permission	created_at/updated_at	timestamp	Audit timestamps	Required	System
Integration	id	UUID/string	Unique record identifier	Required	System
Integration	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Integration	created_at/updated_at	timestamp	Audit timestamps	Required	System
Collector	id	UUID/string	Unique record identifier	Required	System
Collector	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Collector	created_at/updated_at	timestamp	Audit timestamps	Required	System
Raw Event	id	UUID/string	Unique record identifier	Required	System
Raw Event	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Raw Event	created_at/updated_at	timestamp	Audit timestamps	Required	System
Normalized Event	id	UUID/string	Unique record identifier	Required	System
Normalized Event	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Normalized Event	created_at/updated_at	timestamp	Audit timestamps	Required	System
Alert	id	UUID/string	Unique record identifier	Required	System
Alert	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Alert	created_at/updated_at	timestamp	Audit timestamps	Required	System
Incident/Case	id	UUID/string	Unique record identifier	Required	System
Incident/Case	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Incident/Case	created_at/updated_at	timestamp	Audit timestamps	Required	System
Entity	id	UUID/string	Unique record identifier	Required	System
Entity	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Entity	created_at/updated_at	timestamp	Audit timestamps	Required	System
Asset	id	UUID/string	Unique record identifier	Required	System
Asset	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Asset	created_at/updated_at	timestamp	Audit timestamps	Required	System
Vulnerability	id	UUID/string	Unique record identifier	Required	System
Vulnerability	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Vulnerability	created_at/updated_at	timestamp	Audit timestamps	Required	System
Detection Rule	id	UUID/string	Unique record identifier	Required	System
Detection Rule	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Detection Rule	created_at/updated_at	timestamp	Audit timestamps	Required	System
Playbook	id	UUID/string	Unique record identifier	Required	System
Playbook	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Playbook	created_at/updated_at	timestamp	Audit timestamps	Required	System
Threat Intelligence Object	id	UUID/string	Unique record identifier	Required	System
Threat Intelligence Object	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Threat Intelligence Object	created_at/updated_at	timestamp	Audit timestamps	Required	System
Evidence Item	id	UUID/string	Unique record identifier	Required	System
Evidence Item	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Evidence Item	created_at/updated_at	timestamp	Audit timestamps	Required	System
Report	id	UUID/string	Unique record identifier	Required	System
Report	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Report	created_at/updated_at	timestamp	Audit timestamps	Required	System
Audit Log	id	UUID/string	Unique record identifier	Required	System
Audit Log	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Audit Log	created_at/updated_at	timestamp	Audit timestamps	Required	System
Billing Usage Record	id	UUID/string	Unique record identifier	Required	System
Billing Usage Record	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Billing Usage Record	created_at/updated_at	timestamp	Audit timestamps	Required	System
Compliance Evidence Record	id	UUID/string	Unique record identifier	Required	System
Compliance Evidence Record	tenant_id	UUID/string	Tenant partition identifier	Required except global templates	System
Compliance Evidence Record	created_at/updated_at	timestamp	Audit timestamps	Required	System

Appendix D - Open Decisions for Expert Review

ID	Decision Required
OD-001	Final event analytics backend: OpenSearch vs ClickHouse vs Elastic vs Snowflake/BigQuery per deployment model.
OD-002	Final AI provider/private model strategy per country and regulated customer type.
OD-003	Whether to build proprietary SIEM-lite in MVP or rely on external SIEM for deeper log search initially.

OD-004	Customer-side collector technology and update model.
OD-005	Minimum cyber certifications and accreditations required before selling to regulated sectors.
OD-006	Legal authority-to-act templates and liability caps.
OD-007	Country-specific regulatory packs and data-residency strategy.
OD-008	Pricing and overage model for log volume, storage, AI use, and analyst time.
OD-009	24/7 SOC staffing model: in-house, hybrid, partner, or phased managed operations.
OD-010	Incident response retainer scope and exclusions.
OD-011	Threat intelligence feed selection and budget.
OD-012	Whether to support mobile incident approval app in V2.
OD-013	Target first three client segments and anchor customer onboarding sequence.
OD-014	Third-party penetration test provider and readiness timeline.
OD-015	Data retention defaults by package and jurisdiction.

Appendix E - Bibliography and Source Notes

Source	URL / Note
NIST Cybersecurity Framework 2.0	https://www.nist.gov/cyberframework
NIST CSWP 29, Cybersecurity Framework 2.0 PDF	https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf
NIST SP 800-61 Rev. 2, Computer Security Incident Handling Guide - archived/withdrawn status should be checked before final implementation	https://csrc.nist.gov/pubs/sp/800/61/r2/final
MITRE ATT&CK	https://attack.mitre.org/
OCSF	https://ocsf.io/
Sigma Detection Format	https://sigmahq.io/
Sigma rule documentation	https://sigmahq.io/docs/basics/rules.html
OASIS STIX/TAXII 2.1	https://www.oasis-open.org/standard/taxii-version-2-1/
FIRST TLP	https://www.first.org/tlp/
CIS Controls	https://www.cisecurity.org/controls
Reference customer concept paper	Ghana SOCaS Concept Paper uploaded by user; used only as one potential market/use-case reference, not as full platform scope.

End of SRS. This document is intentionally comprehensive, but the final production version should be validated in workshops with cybersecurity architecture, SOC operations, detection engineering, data engineering, cloud security, legal/compliance, and commercial stakeholders before build execution.

Appendix F - Detailed Screen and UX Requirements

The following screen catalogue is intended to prevent under-scoping of the build. Each screen must be represented in wireframes, user stories, role-permission rules, audit requirements, and acceptance tests before development starts.

ID	Screen	Primary Users	Required Content	Acceptance Notes
SCR-001	Login and MFA	All users	Login, password reset, MFA challenge, SSO entry, session expiry notice, suspicious login warning.	No tenant data rendered before authentication; failed attempts logged; SSO/MFA errors clear.
SCR-002	Tenant Switcher	Internal/MSSP users	Switch authorised tenant context, search tenants, see tenant status and package.	Explicit tenant context visible; support impersonation requires reason and audit.
SCR-003	Customer Home Dashboard	Customer users	Incident summary, risk score, integration health, open tasks, approvals, reports.	Shows only customer-visible content; explains monitoring gaps.
SCR-004	Executive Dashboard	Executive viewer	Board KPIs, major risks, SLA trends, recurring themes, recommendations.	Business language; no raw sensitive logs by default.
SCR-005	SOC Command Queue	SOC analysts	Queue by severity, SLA, tenant, status, data source, assigned analyst, age.	Priority sorting, bulk triage guardrails, no cross-tenant leakage.
SCR-006	Alert Detail	SOC analysts/customer technical users	Alert summary, raw source, normalized fields, entities, enrichment, actions.	Customer view filtered; analyst view includes internal notes.
SCR-007	Incident Case Detail	All case roles	Lifecycle, timeline, evidence, tasks, notes, communications, SLA, approvals.	Full audit trail; status transitions validated.
SCR-008	Investigation Timeline	Analysts/IR	Chronological events, filters, source/ingest times, analyst actions, playbook steps.	Preserve time zones and uncertainty; export controlled.
SCR-009	Entity Graph	Analysts	Users, hosts, IPs, domains, files, cloud resources, vulnerabilities, incidents.	Graph confidence shown; inferred links distinguished.
SCR-010	Evidence Locker	Analysts/compliance	Evidence files, hashes, chain-of-custody, classification, report inclusion.	Immutable metadata; deletion restricted.
SCR-011	SOAR Approval Screen	Customer approvers/SOC managers	Recommended action, impact, evidence, expiry, approval/rejection, comments.	High-impact actions require clear warning and identity confirmation.
SCR-012	Playbook Designer	SOC managers/detection engineers	Triggers, steps, approvals, actions, branches, error handling, simulation.	Versioned drafts; production publish requires approval.
SCR-013	Detection Rule Editor	Detection engineers	Rule code, metadata, mapping, tests, severity, playbook, deployment scope.	Syntax validation, test execution, version history.
SCR-014	Detection Coverage Dashboard	SOC/detection leadership	MITRE coverage, data source dependencies, gaps, noisy rules, FP rates.	Coverage must show prerequisites and limitations.
SCR-015	Threat Intel Console	Threat intel analysts	Indicators, campaigns, confidence, TLP, expiry, sightings, related cases.	Low-confidence indicators cannot auto-block without policy.
SCR-016	Connector Marketplace	Tenant/admin users	Available connectors, setup status, required permissions, package availability.	Unsupported connectors show roadmap/request option.
SCR-017	Connector Setup Wizard	Tenant admins/onboarding	Auth, scopes, validation, test event, mapping confirmation, health.	Least privilege instructions and permission risk displayed.
SCR-018	Integration Health Dashboard	SOC/onboarding/customer	Last event, lag, errors, volume, credential expiry, data-source priority.	Critical failures escalate and show monitoring gap.
SCR-019	Asset Inventory	Customer/SOC	Assets, owners, criticality, tags, vulnerabilities, incidents, source confidence.	Bulk edit guarded; stale assets flagged.
SCR-020	Identity Inventory	Customer/SOC	Users, privileged accounts, MFA, groups, risky sign-ins, incidents.	Sensitive identity data role-restricted.
SCR-021	Vulnerability/Exposure Dashboard	Customer/SOC	Vulns by severity, exploitability, asset criticality, remediation tasks.	Prioritisation explained; customer task creation supported.
SCR-022	Reports Library	Customer/compliance	Generated reports, drafts, schedules, approval status, downloads.	Download logged; report scope visible.
SCR-023	Report Builder	Compliance/customer success	Template, period, incidents, metrics, evidence, commentary, recipients.	Large reports async; exports permission-filtered.
SCR-024	Compliance Mapping	Compliance users	Frameworks, controls, evidence, owners, gaps, exceptions, due dates.	Templates configurable by country/sector.
SCR-025	Tenant Settings	Tenant admins	Profile, contacts, escalation matrix, business hours, data retention, authority-to-act.	Material changes require audit and sometimes approval.
SCR-026	Package & Usage	Customer/internal finance	Entitlements, usage, overage risk, log volume, connectors, assets, reports.	Commercial boundaries clear; internal margin view hidden from customer.
SCR-027	MSSP Partner Console	Partner admins	Downstream tenants, branding, analyst allocation, package templates, partner reports.	Partner users cannot access platform global admin controls.
SCR-028	Platform Admin Console	Super admins	Global settings, feature flags, content packs, tenants, system status.	Privileged actions require strong audit.
SCR-029	Audit Log Search	Compliance/security	User actions, system actions, AI, exports, admin, SOAR actions.	Tamper-evident, filterable, export controlled.
SCR-030	AI Copilot Panel	Analysts/customer success	Summary, recommended next steps, report draft, query assistant, confidence.	Evidence-linked outputs; unsafe recommendations reportable.
SCR-031	Major Incident War Room	Incident responders	Roles, live timeline, comms, tasks, approvals, customer updates, decision log.	P1 mode with escalation and communication discipline.
SCR-032	Customer Onboarding Workspace	Onboarding team	Checklist, data sources, test alerts, contacts, assets, baselines, go-live.	No production monitoring until mandatory checklist complete.
SCR-033	Knowledge Base	SOC/customer	Playbooks, advisories, customer guidance, analyst notes, FAQs.	Versioned, searchable, customer-visible tagging.
SCR-034	Release Notes / Status	All users	Platform status, scheduled maintenance, release notes, known issues.	Customer-visible and internal versions separated.
SCR-035	Support Request Screen	Customer/admin	Platform issue, connector issue, report request, feature request, billing issue.	Support SLA linked to package.

Appendix G - API and Service Interface Requirements

The API catalogue below is a build seed. Final endpoint names may change, but the product must expose equivalent capabilities with authentication, authorization, validation, rate limiting, idempotency for writes where appropriate, error codes, audit logs, and tenant scoping.

ID	Domain	Method	Indicative Endpoint	Purpose	Security/Acceptance
API-TEN-001	Tenant	POST	/tenant/create-tenant	Create tenant	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-002	Tenant	PATCH	/tenant/update-tenant-profile	Update tenant profile	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-003	Tenant	GET	/tenant/get-tenant-settings	Get tenant settings	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-004	Tenant	GET	/tenant/list-tenant-users	List tenant users	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-005	Tenant	PATCH	/tenant/archive-tenant	Archive tenant	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-006	Tenant	POST	/tenant/export-tenant-data	Export tenant data	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-007	Tenant	PATCH	/tenant/set-package	Set package	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-008	Tenant	PATCH	/tenant/set-data-residency	Set data residency	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-009	Tenant	PATCH	/tenant/set-retention-policy	Set retention policy	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-TEN-010	Tenant	GET	/tenant/get-tenant-health	Get tenant health	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-001	Identity	PATCH	/identity/invite-user	Invite user	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-002	Identity	PATCH	/identity/deactivate-user	Deactivate user	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-003	Identity	PATCH	/identity/assign-role	Assign role	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-004	Identity	PATCH	/identity/remove-role	Remove role	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-005	Identity	GET	/identity/get-access-review	Get access review	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-006	Identity	POST	/identity/create-api-token	Create API token	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-007	Identity	POST	/identity/rotate-api-token	Rotate API token	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-008	Identity	PATCH	/identity/revoke-api-token	Revoke API token	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-009	Identity	PATCH	/identity/start-sso-config	Start SSO config	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-IDE-010	Identity	PATCH	/identity/validate-sso-config	Validate SSO config	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-001	Ingestion	PATCH	/ingestion/post-alert	Post alert	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-002	Ingestion	PATCH	/ingestion/post-event-batch	Post event batch	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-003	Ingestion	PATCH	/ingestion/post-webhook-event	Post webhook event	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-004	Ingestion	PATCH	/ingestion/register-collector	Register collector	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-005	Ingestion	GET	/ingestion/get-collector-config	Get collector config	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-006	Ingestion	PATCH	/ingestion/update-collector-health	Update collector health	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-007	Ingestion	GET	/ingestion/get-ingestion-status	Get ingestion status	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-008	Ingestion	PATCH	/ingestion/replay-event-batch	Replay event batch	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-009	Ingestion	GET	/ingestion/get-parser-errors	Get parser errors	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ING-010	Ingestion	PATCH	/ingestion/acknowledge-source-outage	Acknowledge source outage	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-001	Case	POST	/case/create-alert	Create alert	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-002	Case	GET	/case/get-alert	Get alert	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-003	Case	PATCH	/case/update-alert-disposition	Update alert disposition	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-004	Case	POST	/case/create-incident	Create incident	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-005	Case	PATCH	/case/update-incident	Update incident	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-006	Case	PATCH	/case/assign-case	Assign case	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-007	Case	PATCH	/case/add-note	Add note	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-008	Case	PATCH	/case/add-task	Add task	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-009	Case	PATCH	/case/attach-evidence	Attach evidence	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-CAS-010	Case	PATCH	/case/close-incident	Close incident	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-001	Investigation	PATCH	/investigation/search-events	Search events	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-002	Investigation	GET	/investigation/get-entity-profile	Get entity profile	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-003	Investigation	GET	/investigation/get-entity-graph	Get entity graph	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-004	Investigation	GET	/investigation/get-timeline	Get timeline	Tenant scoped; RBAC/ABAC enforced; audit where material.

Confidential - Draft for build planning and expert validation

Best-in-Class Cyber Security Operations Platform - Full SRS

					audit where material.
API-INV-005	Investigation	PATCH	/investigation/save-query	Save query	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-006	Investigation	POST	/investigation/run-hunt-query	Run hunt query	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-007	Investigation	POST	/investigation/export-evidence-subset	Export evidence subset	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-008	Investigation	POST	/investigation/create-finding	Create finding	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-009	Investigation	PATCH	/investigation/link-related-case	Link related case	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-INV-010	Investigation	GET	/investigation/get-raw-event	Get raw event	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-001	Detection	POST	/detection/create-rule	Create rule	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-002	Detection	PATCH	/detection/test-rule	Test rule	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-003	Detection	PATCH	/detection/review-rule	Review rule	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-004	Detection	PATCH	/detection/deploy-rule	Deploy rule	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-005	Detection	PATCH	/detection/rollback-rule	Rollback rule	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-006	Detection	GET	/detection/get-rule-coverage	Get rule coverage	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-007	Detection	GET	/detection/list-false-positives	List false positives	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-008	Detection	PATCH	/detection/submit-tuning-feedback	Submit tuning feedback	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-009	Detection	PATCH	/detection/retire-rule	Retire rule	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-DET-010	Detection	POST	/detection/export-rule-pack	Export rule pack	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-001	Playbook	POST	/playbook/create-playbook	Create playbook	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-002	Playbook	PATCH	/playbook/simulate-playbook	Simulate playbook	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-003	Playbook	PATCH	/playbook/publish-playbook	Publish playbook	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-004	Playbook	POST	/playbook/execute-playbook	Execute playbook	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-005	Playbook	POST	/playbook/approve-action	Approve action	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-006	Playbook	POST	/playbook/reject-action	Reject action	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-007	Playbook	GET	/playbook/get-execution-log	Get execution log	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-008	Playbook	PATCH	/playbook/cancel-execution	Cancel execution	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-009	Playbook	PATCH	/playbook/rollback-guidance	Rollback guidance	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-PLA-010	Playbook	GET	/playbook/list-action-catalog	List action catalog	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-001	AI	POST	/ai/summarise-alert	Summarise alert	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-002	AI	POST	/ai/summarise-incident	Summarise incident	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-003	AI	POST	/ai/draft-customer-update	Draft customer update	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-004	AI	POST	/ai/draft-pir	Draft PIR	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-005	AI	PATCH	/ai/suggest-rule-tuning	Suggest rule tuning	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-006	AI	POST	/ai/generate-query	Generate query	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-007	AI	PATCH	/ai/explain-threat-intel	Explain threat intel	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-008	AI	PATCH	/ai/evaluate-output-feedback	Evaluate output feedback	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-009	AI	GET	/ai/get-prompt-log	Get prompt log	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-AI-010	AI	PATCH	/ai/set-ai-policy	Set AI policy	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-001	Reporting	POST	/reporting/create-report-draft	Create report draft	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-002	Reporting	POST	/reporting/generate-report	Generate report	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-003	Reporting	POST	/reporting/approve-report	Approve report	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-004	Reporting	PATCH	/reporting/download-report	Download report	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-005	Reporting	PATCH	/reporting/schedule-report	Schedule report	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-006	Reporting	GET	/reporting/list-report-templates	List report templates	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-007	Reporting	POST	/reporting/create-evidence-pack	Create evidence pack	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-008	Reporting	POST	/reporting/export-csv	Export CSV	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-009	Reporting	GET	/reporting/get-dashboard-metrics	Get dashboard metrics	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-REP-010	Reporting	PATCH	/reporting/revoke-report-link	Revoke report link	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-001	Compliance	POST	/compliance/create-framework	Create framework	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-002	Compliance	PATCH	/compliance/map-control	Map control	Tenant scoped; RBAC/ABAC enforced; audit where material.

Confidential - Draft for build planning and expert validation

Best-in-Class Cyber Security Operations Platform - Full SRS

API-COM-003	Compliance	PATCH	/compliance/attach-evidence	Attach evidence	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-004	Compliance	POST	/compliance/create-exception	Create exception	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-005	Compliance	POST	/compliance/approve-risk-acceptance	Approve risk acceptance	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-006	Compliance	GET	/compliance/get-evidence-freshness	Get evidence freshness	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-007	Compliance	POST	/compliance/export-audit-pack	Export audit pack	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-008	Compliance	GET	/compliance/list-regulatory-reports	List regulatory reports	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-009	Compliance	PATCH	/compliance/update-due-date	Update due date	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-COM-010	Compliance	PATCH	/compliance/close-audit-request	Close audit request	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-001	Billing	GET	/billing/get-usage	Get usage	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-002	Billing	GET	/billing/get-entitlements	Get entitlements	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-003	Billing	PATCH	/billing/update-entitlement	Update entitlement	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-004	Billing	POST	/billing/create-overage-alert	Create overage alert	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-005	Billing	POST	/billing/export-billing-usage	Export billing usage	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-006	Billing	POST	/billing/record-service-hours	Record service hours	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-007	Billing	POST	/billing/create-addon-service	Create addon service	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-008	Billing	PATCH	/billing/set-framework-minimum	Set framework minimum	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-009	Billing	GET	/billing/get-margin-view	Get margin view	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-BIL-010	Billing	PATCH	/billing/suspend-service	Suspend service	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-001	Admin	GET	/admin/list-tenants	List tenants	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-002	Admin	GET	/admin/get-system-health	Get system health	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-003	Admin	PATCH	/admin/set-feature-flag	Set feature flag	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-004	Admin	PATCH	/admin/manage-content-pack	Manage content pack	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-005	Admin	POST	/admin/run-data-repair-job	Run data repair job	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-006	Admin	GET	/admin/get-audit-log	Get audit log	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-007	Admin	PATCH	/admin/set-global-policy	Set global policy	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-008	Admin	POST	/admin/rotate-platform-key	Rotate platform key	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-009	Admin	POST	/admin/create-maintenance-banner	Create maintenance banner	Tenant scoped; RBAC/ABAC enforced; audit where material.
API-ADM-010	Admin	GET	/admin/get-release-status	Get release status	Tenant scoped; RBAC/ABAC enforced; audit where material.

Appendix H - Data Model Table Catalogue

The table catalogue below provides enough structure for the initial database design and domain model. Final physical schemas may vary by data store, but logical ownership, retention, tenant scope, and security classification must be preserved.

Table	Purpose	Key Fields	Retention	Classification
tenants	Customer/partner/deployment partition	tenant_id, parent_tenant_id, name, status, package_id, region, data_residency, created_at	Permanent + retention rules	High
tenant_settings	Tenant configuration	tenant_id, escalation_matrix, business_hours, authority_policy, retention_policy	Life of tenant	High
users	Platform users	user_id, tenant_id, email, name, status, mfa_status, last_login	Life of account + audit retention	High
roles	Role definitions	role_id, name, scope, description	Permanent	Medium
user_roles	User role assignments	user_id, role_id, tenant_id, assigned_by, assigned_at	Life + audit	High
api_tokens	Machine credentials metadata	token_id, tenant_id, scopes, expiry, status, last_used	Until revoked + audit	High
integrations	Connector configuration metadata	integration_id, tenant_id, type, status, scopes, health, last_success	Life of connector	High
connector_secrets	Encrypted references only	secret_ref, integration_id, vault_path, rotation_due	Life of connector	Restricted
collectors	Collector inventory	collector_id, tenant_id, version, host, status, last_seen	Life of collector	High
raw_events	Raw payload metadata	raw_id, tenant_id, source, object_ref, hash, ingest_time	Policy based	Restricted
normalized_events	Canonical events	event_id, tenant_id, source, event_time, category, activity, severity, entity_refs	Hot/warm retention	High
alerts	Security alerts	alert_id, tenant_id, severity, status, rule_id, source, risk_score	Case retention	High
incidents	Cases/incidents	incident_id, tenant_id, severity, status, owner, sla_due, disposition	Case retention	High
incident_alerts	Alert-case links	incident_id, alert_id, link_type	Case retention	High
incident_timeline	Timeline entries	entry_id, incident_id, time, type, summary, source_ref	Case retention	High
case_notes	Notes	note_id, incident_id, author, visibility, body, created_at	Case retention	High
case_tasks	Tasks	task_id, incident_id, assignee, status, due_at, outcome	Case retention	Medium
evidence_items	Evidence artifacts	evidence_id, incident_id, object_ref, hash, classification, custody	Legal/evidence retention	Restricted
assets	Asset inventory	asset_id, tenant_id, type, hostname, owner, criticality, confidence	Life of asset + history	High
identities	Identity inventory	identity_id, tenant_id, upn, privilege, mfa, status, risk	Policy based	High
vulnerabilities	Vulnerability findings	vuln_id, tenant_id, cve, severity, exploitability, status	Policy based	Medium
asset_vulnerabilities	Asset-vulnerability mapping	asset_id, vuln_id, first_seen, last_seen, status	Policy based	Medium
detection_rules	Rules	rule_id, version, status, severity, logic_ref, owner, mitre	Permanent + versions	Medium
rule_tests	Rule tests	test_id, rule_id, input_ref, expected, result	Permanent	Medium
rule_feedback	FP/tuning feedback	feedback_id, rule_id, alert_id, reason, submitted_by	Rule lifecycle	Medium
playbooks	Playbook definitions	playbook_id, version, status, trigger, owner	Permanent + versions	Medium
playbook_executions	Run logs	execution_id, playbook_id, incident_id, status, started_at, ended_at	Audit retention	High
approval_requests	Action approvals	approval_id, action, risk_class, approver, status, expiry	Audit retention	High
action_catalog	Allowed response actions	action_id, class, connector, permissions, rollback_guidance	Permanent	Medium
threat_intel_objects	Intel objects	intel_id, type, value, source, confidence, tlp, expiry	Intel retention	Medium
intel_sightings	Indicator sightings	sighting_id, intel_id, tenant_id, event_ref, first_seen	Policy based	Medium
reports	Reports metadata	report_id, tenant_id, template, status, period, object_ref	Report retention	High
report_downloads	Download audit	download_id, report_id, user_id, time, ip	Audit retention	High
compliance_controls	Controls	control_id, framework, requirement, owner	Permanent	Medium
control_mappings	Evidence mappings	control_id, evidence_id, incident_id, status	Compliance retention	Medium
audit_logs	Audit events	audit_id, tenant_id, actor, action, target, time, result	Immutable retention	Restricted
usage_records	Billing/metering	usage_id, tenant_id, metric, quantity, period	Finance retention	Medium
service_hours	Professional services tracking	record_id, tenant_id, service_type, hours, case_ref	Finance retention	Medium
feature_flags	Feature configuration	flag_id, scope, value, rollout, owner	Permanent + history	Medium
system_health	Platform health metrics	metric_id, service, status, latency, errors, time	Operational retention	Low

Appendix I - Detailed Connector Specifications

Connector	Inbound Data	Outbound Actions	Authentication	Implementation Notes
Microsoft 365	Audit logs, email events, mailbox rules, user activity, security alerts.	Quarantine email, search mailbox, create investigation task where permissions allow.	OAuth app registration with least-privilege Graph/API permissions.	High MVP priority due to common adoption and phishing/BEC value.
Entra ID	Sign-ins, risky users, group changes, privilege changes, MFA status, app consent.	Disable user, revoke sessions, require password reset, remove risky app consent where allowed.	OAuth / Graph permissions; privileged actions separated from read-only ingestion.	Identity is core to modern attacks.
Microsoft Defender	Endpoint alerts, device info, vulnerabilities, investigation package, incidents.	Isolate device, collect package, run AV scan where pre-authorised.	OAuth security API permissions; action permissions separately gated.	Primary early EDR connector.
Microsoft Sentinel	Incidents, alerts, analytics rule outputs, log query handoff.	Create/update incident, sync comments, run approved query if configured.	Azure app registration and workspace permissions.	Useful where customer already has SIEM investment.
CrowdStrike Falcon	Detections, hosts, users, behaviours, incidents.	Contain host, lift containment, add IOC where authorised.	OAuth client credentials; action scopes separately configured.	Advanced MDR tier.
SentinelOne	Threats, agents, endpoints, alerts.	Isolate endpoint, kill/quarantine where authorised.	API token/OAuth depending on deployment.	Advanced MDR tier.
Sophos	Endpoint alerts, server/workload alerts, email/network events where available.	Endpoint isolation/remediation where supported.	API credentials and tenant-specific scopes.	SME/mid-market value.
Fortinet	Firewall/syslog, FortiAnalyzer alerts, VPN logs, blocked traffic.	Block IP/domain, update address group where authorised.	Syslog/API; TLS and IP allowlisting.	Network monitoring core.
Palo Alto	Firewall traffic/threat logs, Prisma/Cortex alerts where licensed.	Block indicators, update security policies where authorised.	API key/OAuth; policy action carefully controlled.	Enterprise/critical tier.
Cisco	Meraki, Umbrella, Secure Endpoint, firewall/network alerts depending on product.	Block domain/IP, create incident, update policy where authorised.	Product-specific APIs.	Phase 3 due to product family breadth.
Okta	Sign-ins, user lifecycle, MFA events, app assignments, suspicious activity.	Suspend user, reset factors, revoke sessions where authorised.	API token/OAuth with least privilege.	Important for enterprises outside Microsoft identity.
Google Workspace	Admin audit, login audit, Gmail audit, Drive activity, alert center.	Suspend user, revoke token, investigate email where authorised.	OAuth service account/domain-wide delegation.	Important for SME/startup/public sector using Google.
AWS	CloudTrail, GuardDuty, IAM, Config, Security Hub, EC2/S3/Lambda context.	Disable key, apply security group change, tag resource, create ticket.	Cross-account role with external ID and least privilege.	Cloud tier.
Azure	Activity logs, Defender for Cloud, IAM, resource inventory.	Disable credential, update NSG, tag resource, create remediation task.	Azure app/managed identity permissions.	Cloud tier and Microsoft ecosystem.
GCP	Cloud Audit Logs, Security Command Center, IAM, asset inventory.	Disable service account key, update firewall, create ticket.	Service account with least privilege.	Phase 3 cloud.
Cloudflare	DNS, WAF, Access, Zero Trust logs, blocked traffic.	Block IP/domain, update WAF list, create Zero Trust action.	API token with scoped permissions.	Edge and web app monitoring.
Zscaler	Web access logs, private access, cloud firewall events.	Block URL/category, create ticket; automated changes cautiously gated.	API credentials / log streaming.	SASE tier.
ServiceNow	Incidents, tasks, change context, CMDB.	Create/update tickets, sync status/comments, link cases.	OAuth/basic/service account.	Enterprise workflow sync.
Jira	Issues, tasks, project workflows.	Create/update issues, sync comments/status.	OAuth/API token.	Useful for tech teams and remediation.
Slack	Notifications, approvals, war-room channels.	Send message, request approval, create channel where allowed.	OAuth bot app.	Communication channel.
Teams	Notifications, approvals, war-room channels.	Send message/card, request approval, create channel where allowed.	Graph app permissions.	Communication channel.
Vulnerability Scanner	Findings, assets, severity, exploitability, scan status.	Create remediation task, update exception.	API credentials per vendor.	Exposure context and add-on service.
Syslog	Network/device/server logs from many products.	Inbound only in MVP; response through specific connector or manual task.	TLS syslog, collector certificate, parsing profile.	Broadest initial coverage.
Customer API	Sector-specific events and custom systems.	Custom actions by SOW only.	Mutual TLS/API token/OAuth.	Requires specification and professional services.

Appendix J - Detailed Playbook Step Templates

Step ID	Playbook	Step	Inputs	Outputs/Acceptance
PB-001-01	Phishing	Trigger and classify	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-02	Phishing	Gather source alert and raw evidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-03	Phishing	Enrich entities and indicators	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-04	Phishing	Check asset/user criticality	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-05	Phishing	Search related events	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-06	Phishing	Assess severity and confidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-07	Phishing	Determine customer notification need	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-08	Phishing	Run safe containment recommendation	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-09	Phishing	Request approval if required	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-10	Phishing	Execute authorised action	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-11	Phishing	Create customer/internal tasks	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-12	Phishing	Monitor for recurrence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-13	Phishing	Close with disposition	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-14	Phishing	Capture detection feedback	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-001-15	Phishing	Generate report/PIR if required	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-01	Business Email Compromise	Trigger and classify	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-02	Business Email Compromise	Gather source alert and raw evidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-03	Business Email Compromise	Enrich entities and indicators	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-04	Business Email Compromise	Check asset/user criticality	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-05	Business Email Compromise	Search related events	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-06	Business Email Compromise	Assess severity and confidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-07	Business Email Compromise	Determine customer notification need	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-08	Business Email Compromise	Run safe containment recommendation	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-09	Business Email Compromise	Request approval if required	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-10	Business Email Compromise	Execute authorised action	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-11	Business Email Compromise	Create customer/internal tasks	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-12	Business Email Compromise	Monitor for recurrence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-13	Business Email Compromise	Close with disposition	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-14	Business Email Compromise	Capture detection feedback	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-002-15	Business Email Compromise	Generate report/PIR if required	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-01	Ransomware	Trigger and classify	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-02	Ransomware	Gather source alert and raw evidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-03	Ransomware	Enrich entities and indicators	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-04	Ransomware	Check asset/user criticality	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-05	Ransomware	Search related events	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-06	Ransomware	Assess severity and confidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-07	Ransomware	Determine customer notification need	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-08	Ransomware	Run safe containment recommendation	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-09	Ransomware	Request approval if required	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-10	Ransomware	Execute authorised action	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-11	Ransomware	Create customer/internal tasks	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-12	Ransomware	Monitor for recurrence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-13	Ransomware	Close with disposition	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-14	Ransomware	Capture detection feedback	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-003-15	Ransomware	Generate report/PIR if required	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-004-01	Endpoint Malware	Trigger and classify	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-004-02	Endpoint Malware	Gather source alert and raw evidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-004-03	Endpoint Malware	Enrich entities and indicators	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-004-04	Endpoint Malware	Check asset/user criticality	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-004-05	Endpoint Malware	Search related events	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-004-06	Endpoint Malware	Assess severity and confidence	Inputs from case, connector, rule, tenant policy and analyst	Output logged to case timeline and audit

Best-in-Class Cyber Security Operations Platform - Full SRS

[illegible]

Confidential - Draft for build planning and expert validation

Best-in-Class Cyber Security Operations Platform - Full SRS

[illegible]

Confidential - Draft for build planning and expert validation

Best-in-Class Cyber Security Operations Platform - Full SRS

[illegible]

Confidential - Draft for build planning and expert validation

Best-in-Class Cyber Security Operations Platform - Full SRS

			context.	trail.
PB-015-01	Connector Health Failure	Trigger and classify	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-02	Connector Health Failure	Gather source alert and raw evidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-03	Connector Health Failure	Enrich entities and indicators	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-04	Connector Health Failure	Check asset/user criticality	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-05	Connector Health Failure	Search related events	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-06	Connector Health Failure	Assess severity and confidence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-07	Connector Health Failure	Determine customer notification need	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-08	Connector Health Failure	Run safe containment recommendation	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-09	Connector Health Failure	Request approval if required	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-10	Connector Health Failure	Execute authorised action	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-11	Connector Health Failure	Create customer/internal tasks	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-12	Connector Health Failure	Monitor for recurrence	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-13	Connector Health Failure	Close with disposition	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-14	Connector Health Failure	Capture detection feedback	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.
PB-015-15	Connector Health Failure	Generate report/PIR if required	Inputs from case, connector, rule, tenant policy and analyst context.	Output logged to case timeline and audit trail.

Appendix K - Detailed Build User Stories and Acceptance Criteria

[illegible]

Best-in-Class Cyber Security Operations Platform - Full SRS

			clearly.	
US-027	Threat intel enrichment	As an authorised user, I want threat intel enrichment capability so that the SOC platform can deliver secure and measurable operations.	Given valid role and tenant context, when the user completes the workflow, then the system updates records, enforces permissions, logs audit events, and displays success/failure clearly.	Must include negative permission test, tenant isolation test, audit log test, and error path test.
US-028	Vulnerability context	As an authorised user, I want vulnerability context capability so that the SOC platform can deliver secure and measurable operations.	Given valid role and tenant context, when the user completes the workflow, then the system updates records, enforces permissions, logs audit events, and displays success/failure clearly.	Must include negative permission test, tenant isolation test, audit log test, and error path test.
US-029	Customer task workflow	As an authorised user, I want customer task workflow capability so that the SOC platform can deliver secure and measurable operations.	Given valid role and tenant context, when the user completes the workflow, then the system updates records, enforces permissions, logs audit events, and displays success/failure clearly.	Must include negative permission test, tenant isolation test, audit log test, and error path test.
US-030	Offboarding archive	As an authorised user, I want offboarding archive capability so that the SOC platform can deliver secure and measurable operations.	Given valid role and tenant context, when the user completes the workflow, then the system updates records, enforces permissions, logs audit events, and displays success/failure clearly.	Must include negative permission test, tenant isolation test, audit log test, and error path test.

Appendix L - Operational Reports and KPI Definitions

KPI	Definition	Segmentation / Notes
Mean Time to Triage	Time from alert creation/ingestion to first analyst triage decision.	By tenant, severity, data source, analyst shift.
Mean Time to Acknowledge	Time from customer-visible incident to customer acknowledgement or SOC acknowledgement depending on SLA.	By package and severity.
Mean Time to Containment	Time from confirmed incident to approved containment action completion.	Only for incidents where containment is applicable.
False Positive Rate	Percentage of alerts closed as false positive by rule/source/tenant.	Measured after onboarding tuning window.
Data Source Health	Percentage of time priority data sources are active and within lag threshold.	Critical source outage affects service review.
SLA Compliance	Incidents handled within contractual thresholds.	By severity and package.
Analyst Utilisation	Queue work, cases, investigations, service tasks, admin time.	SOC management only.
Detection Coverage	Mapped ATT&CK techniques with active rules and required data sources.	Shows limitations where data is missing.
Response Action Success	Approved playbook actions completed successfully.	By connector/action/risk class.
Customer Remediation Age	Open customer tasks by age and severity.	Used in service reviews.
Evidence Freshness	Age of evidence for compliance controls.	Compliance dashboard.
Churn/Renewal Risk	Combination of usage, SLA, incidents, satisfaction, receivables.	Commercial dashboard.
Gross Margin by Tenant	Revenue less tooling, storage, analyst time, support, pro services.	Internal finance only.
AI Assistance Acceptance	AI suggestions accepted/edited/rejected/unsafe.	AI governance.
Connector Error Rate	Failed connector runs versus total runs.	Platform health and onboarding quality.

Appendix M - Final Truth Notes and Build Warnings

- Building the platform is achievable with Developers and a disciplined engineering team, but SOC credibility depends on expert configuration, detection engineering, analyst operations, incident response governance, and customer trust.
- The system must be positioned honestly: it reduces risk and improves detection/response; it cannot guarantee breach prevention.
- Commercial packages must enforce limits, or anchor customers can destroy margin through bespoke scope creep.
- AI must be designed with safety controls, evidence grounding, tenant isolation, and human approval for high-impact actions.
- The first commercial release should be narrow enough to operate well, not broad enough to look impressive but fail in production.
- The platform should not try to replace every SIEM/EDR/firewall. It should provide a strong SOC operating layer above them, with native capabilities where strategically valuable.
- A senior cybersecurity architect and SOC operations lead should review this SRS before using it as the final source of truth for implementation.